Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: S&S Dairy, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
348 E Monte Vista RD	Ceres	Stanislaus	95307

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1968

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X026-X014-XXXX

B. OPERATORS

Bylsma, Darrin

Operator name: Bylsma, Darrin

Telephone no.: Landline: ; Cellular: (209) 606-4894

Mailing Address Number and Street	City	State	Zip Code
348 E Monte Vista AVE	Ceres	CA	95307

Inc., S & S Dairy

Operator name: Inc., S & S Dairy

Telephone no.: Landline: ; Cellular: (209) 606-4894

Mailing Address Number and Street	City	State	Zip Code
5870 Crowslanding RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

Bylsma, Darrin

Legal owner name: Bylsma, Darrin

Telephone no.: Landline: ; Cellular: (209) 606-4894

Mailing Address Number and Street	City	State	Zip Code
348 E Monte Vista AVE	Ceres	CA	95307

Bylsma, Sam

Legal owner name: Bylsma, Sam		Telephone no.:	(209) 537-6336
		Landline	Cellular
5870 Crows Landing	Crows Landing	CA	95307
Mailing Address Number and Street	City	State	Zip Code

Limited Partnership, Hoffman

Legal owner name: Limited Partnership, Hoffman		Telephone no.:	(209) 606-4894
		Landline	Cellular
5870 Crowslanding RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	450	465	300	280
Number under roof	1,950	290	0	0	0	0
Maximum number	1,950	290	450	465	300	280
Average number	1,950	290	450	465	300	280
Avg live weight (lbs)	1,400	1,450	1,000	800		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 66,449.13 *tons per reporting period*

Total nitrogen from manure: 835,356.48 *lbs per reporting period*

After ammonia losses (30% loss applied): 584,749.54 *lbs per reporting period*

Total phosphorus from manure: 136,694.34 *lbs per reporting period*

Total potassium from manure: 378,567.59 *lbs per reporting period*

Total salt from manure: 984,843.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 39,361,400 *gallons*

Total nitrogen generated: 127,283.88 *lbs*

Total phosphorus generated: 17,884.18 *lbs*

Total potassium generated: 231,675.13 *lbs*

Total salt generated: 1,190,362.51 *lbs*

	37,449,000	*gallons applied*
+	1,912,400	*gallons exported*
-	0	*gallons imported*
=	39,361,400	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well #1	Ground water
Domestic Well - Well #3	Ground water
Domestic Well - Well #4	Ground water
Domestic Well - Well #5	Ground water
Domestic Well - Well #7	Ground water

Source Description	Type
Domestic Well - Well #8	Ground water
Domestic Well - Well #9	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
05/22/2023	Process wastewater	800,000.00 *gal*	477.00	375.00	0.00	0.80	59.70	854.00		4,040
08/21/2023	Process wastewater	1,112,400.00 *gal*	178.00	153.00	0.00	0.40	38.00	400.00		2,700

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	4,845.88	751.31	9,414.50	52,035.08
Total exports for all materials	4,845.88	751.31	9,414.50	52,035.08

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	35	35	2	process wastewater	X022-X009-X004-XXXX X022-X009-X005-XXXX X022-X009-X009-XXXX
10	40	40	0	none	X022-X035-X011-XXXX
11	40	40	6	manure	X058-X025-X001-XXXX
12-Brad	35	35	6	manure	X057-X002-X003-XXXX X057-X002-X004-XXXX X057-X002-X006-XXXX
12-NE	35	35	2	manure	X057-X002-X003-XXXX
12-NW	35	35	2	manure	X057-X002-X003-XXXX
12-SE	35	35	2	manure	X057-X002-X003-XXXX
12-SW	45	45	2	manure	X057-X002-X003-XXXX
13-E	40	40	2	manure	X057-X005-X005-XXXX
13-N	40	40	2	manure	X057-X005-X005-XXXX
13-S	40	40	2	manure	X057-X005-X005-XXXX
14	15	15	6	process wastewater	X022-X008-X008-XXXX
16	10	10	2	manure	X022-X028-X011-XXXX
17	13	13	6	manure	X022-X008-X029-XXXX
2	35	35	2	process wastewater	X022-X009-X006-XXXX X022-X009-X007-XXXX
3	40	40	2	process wastewater	X022-X026-X014-XXXX
4	30	30	2	process wastewater	X022-X026-X003-XXXX X022-X026-X014-XXXX
5	55	55	2	process wastewater	X022-X026-X015-XXXX X022-X026-X016-XXXX X022-X026-X017-XXXX X022-X026-X018-XXXX
6	30	30	6	process wastewater	X022-X023-X005-XXXX
7	21	21	2	process wastewater	X022-X027-X013-XXXX X022-X027-X014-XXXX
9	60	60	0	none	X041-X030-X001-XXXX X041-X030-X025-XXXX X041-X030-X030-XXXX

Totals for areas that were used for application	629	629	58		
Totals for areas that were not used for application	100	100	0		
Land application area totals	729	729	58		

B. CROPS AND HARVESTS

1

Field name: 1

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	493.12 *ton*	Dry-weight		69.0	18,500.00	3,500.00	26,500.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.09	161.60	30.57	231.48	873.53

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	988.52 *ton*	Dry-weight		64.3	12,500.00	1,900.00	16,500.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.24	252.07	38.32	332.74	1,129.29

11

Field name: 11

11

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	47.36 *ton*	Dry-weight		8.7	39,800.00	2,200.00	14,800.00		9.50
06/09/2023	50.55 *ton*	Dry-weight		8.7	39,800.00	2,200.00	14,800.00		9.50
07/07/2023	52.48 *ton*	Dry-weight		8.7	39,800.00	2,200.00	14,800.00		9.50
08/05/2023	57.33 *ton*	Dry-weight		7.6	26,900.00	2,400.00	11,400.00		6.30
09/03/2023	51.21 *ton*	Dry-weight		7.6	26,900.00	2,400.00	11,400.00		6.30
10/03/2023	48.14 *ton*	Dry-weight		7.6	26,900.00	2,400.00	11,400.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.68	467.96	32.48	184.13	1,108.24

12-Brad

Field name: 12-Brad

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 35 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	39.86 *ton*	Dry-weight		7.5	32,500.00	2,300.00	12,000.00		6.00
06/08/2023	43.79 *ton*	Dry-weight		7.5	32,500.00	2,300.00	12,000.00		6.00
07/06/2023	49.45 *ton*	Dry-weight		7.5	32,500.00	2,300.00	12,000.00		6.00
08/04/2023	54.77 *ton*	Dry-weight		8.8	17,500.00	1,400.00	8,400.00		6.10
09/02/2023	46.92 *ton*	Dry-weight		8.8	17,500.00	1,400.00	8,400.00		6.10
10/02/2023	37.27 *ton*	Dry-weight		8.8	17,500.00	1,400.00	8,400.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.77	355.38	26.32	145.25	863.87

12-NE

12-NE

Field name: 12-NE

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	450.15 *ton*	Dry-weight		70.7	18,800.00	3,800.00	30,100.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.86	141.69	28.64	226.86	776.29

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,036.25 *ton*	Dry-weight		65.8	13,600.00	2,100.00	16,300.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.61	275.42	42.53	330.10	1,134.07

12-NW

Field name: 12-NW

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	458.31 *ton*	Dry-weight		68.6	17,500.00	3,600.00	28,600.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.09	143.91	29.60	235.19	805.89

12-NW

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,028.33 *ton*	Dry-weight		65.7	13,300.00	1,700.00	16,100.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.38	268.07	34.26	324.50	1,148.85

12-SE

Field name: 12-SE

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	425.45 *ton*	Dry-weight		70.8	18,800.00	3,300.00	27,500.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.16	133.46	23.43	195.22	766.69

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	984.71 *ton*	Dry-weight		64.7	12,800.00	1,900.00	16,100.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.13	254.25	37.74	319.79	1,072.60

12-SW

Field name: 12-SW

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	576.33 *ton*	Dry-weight		73.0	19,800.00	3,600.00	29,500.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.81	136.94	24.90	204.02	740.01

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,268.25 *ton*	Dry-weight		63.6	13,600.00	2,200.00	17,300.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.18	279.04	45.14	354.95	1,169.50

13-E

Field name: 13-E

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	519.34 *ton*	Dry-weight		71.6	19,200.00	3,300.00	26,900.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.98	141.59	24.34	198.38	715.34

13-E

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,141.25 *ton*	Dry-weight		63.8	14,300.00	2,200.00	16,200.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.53	295.39	45.44	334.64	1,094.80

13-N

Field name: 13-N

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	491.87 *ton*	Dry-weight		72.9	20,600.00	3,300.00	25,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.30	137.30	21.99	171.29	706.47

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,179.55 *ton*	Dry-weight		65.9	12,200.00	1,900.00	17,500.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.49	245.36	38.21	351.95	1,166.46

13-S

Field name: 13-S

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	497.56 *ton*	Dry-weight		69.4	18,400.00	3,700.00	28,100.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.44	140.07	28.17	213.92	738.43

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,185.45 *ton*	Dry-weight		65.1	12,500.00	1,800.00	15,500.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.64	258.58	37.23	320.63	1,199.79

14

Field name: 14

14

05/08/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 15 Plant date: 05/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	16.10 *ton*	Dry-weight		8.3	34,600.00	2,300.00	17,000.00		9.30
06/08/2023	18.45 *ton*	Dry-weight		8.3	34,600.00	2,300.00	17,000.00		9.30
07/06/2023	21.34 *ton*	Dry-weight		8.3	34,600.00	2,300.00	17,000.00		9.30
08/04/2023	22.41 *ton*	Dry-weight		7.7	20,600.00	1,400.00	8,900.00		5.80
09/02/2023	20.95 *ton*	Dry-weight		7.7	20,600.00	1,400.00	8,900.00		5.80
10/02/2023	16.88 *ton*	Dry-weight		7.7	20,600.00	1,400.00	8,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.74	389.16	26.10	182.15	1,065.50

16

Field name: 16

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	129.12 *ton*	Dry-weight		69.9	17,500.00	2,800.00	23,400.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.91	136.03	21.76	181.89	722.89

16

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	281.24 *ton*	Dry-weight		64.7	12,000.00	1,900.00	13,900.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.12	238.27	37.73	275.99	933.21

17

Field name: 17

05/08/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 13 Plant date: 05/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	13.25 *ton*	Dry-weight		8.2	36,500.00	2,300.00	18,800.00		9.70
06/08/2023	14.65 *ton*	Dry-weight		8.2	36,500.00	2,300.00	18,800.00		9.70
07/06/2023	15.28 *ton*	Dry-weight		8.2	36,500.00	2,300.00	18,800.00		9.70
08/04/2023	17.45 *ton*	Dry-weight		10.5	26,100.00	2,300.00	20,200.00		8.40
09/02/2023	16.71 *ton*	Dry-weight		10.5	26,100.00	2,300.00	20,200.00		8.40
10/02/2023	13.15 *ton*	Dry-weight		10.5	26,100.00	2,300.00	20,200.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	6.96	392.61	29.01	246.24	1,138.73

2

Field name: 2

2

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	510.11 *ton*	Dry-weight		72.2	20,000.00	4,700.00	30,500.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.57	162.07	38.09	247.16	891.38

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,004.45 *ton*	Dry-weight		66.3	11,900.00	1,500.00	18,200.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.70	230.18	29.01	352.04	1,141.23

3

Field name: 3

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	632.08 *ton*	Dry-weight		71.2	17,000.00	3,200.00	25,500.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	154.73	29.13	232.10	891.99

3

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,310.11 *ton*	Dry-weight		64.4	14,000.00	2,400.00	17,000.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.75	326.48	55.97	396.44	1,375.88

4

Field name: 4

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	428.14 *ton*	Dry-weight		69.6	16,900.00	2,900.00	26,900.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.27	146.64	25.16	233.41	867.70

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	842.47 *ton*	Dry-weight		64.2	13,800.00	2,100.00	15,600.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.08	277.48	42.22	313.67	1,065.67

5

Field name: 5

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	715.64 *ton*	Dry-weight		71.5	19,700.00	4,100.00	34,600.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.01	146.11	30.41	256.62	734.25

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,643.74 *ton*	Dry-weight		64.6	13,000.00	1,900.00	17,300.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.89	275.07	40.20	366.06	1,184.93

6

Field name: 6

6

04/30/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 30 Plant date: 04/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	34.44 *ton*	Dry-weight		8.5	45,400.00	2,900.00	16,500.00		11.10
06/09/2023	40.23 *ton*	Dry-weight		8.5	45,400.00	2,900.00	16,500.00		11.10
07/07/2023	44.77 *ton*	Dry-weight		7.9	15,500.00	1,100.00	6,900.00		4.70
08/05/2023	46.00 *ton*	Dry-weight		7.9	15,500.00	1,100.00	6,900.00		4.70
09/03/2023	39.64 *ton*	Dry-weight		7.9	15,500.00	1,100.00	6,900.00		4.70
10/03/2023	34.15 *ton*	Dry-weight		7.9	15,500.00	1,100.00	6,900.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.97	363.40	24.32	144.87	980.48

7

Field name: 7

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	250.88 *ton*	Dry-weight		71.3	19,600.00	3,400.00	28,600.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.95	134.40	23.32	196.12	678.88

7

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	603.14 *ton*	Dry-weight		64.2	13,600.00	2,500.00	15,200.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.72	279.67	51.41	312.58	1,089.90

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/02/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.85	13.57	161.80	865.98	1,062,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.31	5,994,150.00 *gal*
Application event totals		92.85	13.57	161.80	920.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	118.93	14.86	212.58	1,005.63	1,044,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	55.53	6,128,850.00 *gal*
Application event totals		118.93	14.86	212.58	1,061.16	

1 - 06/03/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.80	4,613,475.00 *gal*
Application event totals		0.00	0.00	0.00	41.80	

1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.49	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	41.49	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.19	4,546,125.00 *gal*
Application event totals		0.00	0.00	0.00	41.19	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	136.36	17.04	243.73	1,153.01	1,197,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.49	4,579,800.00 *gal*
Application event totals		136.36	17.04	243.73	1,194.51	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	129.19	16.14	230.90	1,092.33	1,134,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.49	4,579,800.00 *gal*
Application event totals		129.19	16.14	230.90	1,133.82	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.00	10.44	109.87	741.61	1,152,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.02	4,748,175.00 *gal*
Application event totals		49.00	10.44	109.87	784.63	

1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.10	4,647,150.00 *gal*
Application event totals		0.00	0.00	0.00	42.10	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,714,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

11 - 11/15/2020: Alfalfa, hay

Field name: 11

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	317.45	135.47	281.06	0.00	1,200.00 *ton*
Application event totals		317.45	135.47	281.06	0.00	

04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.46	6,364,575.00 *gal*
Application event totals		0.00	0.00	0.00	50.46	

05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.19	6,330,900.00 *gal*
Application event totals		0.00	0.00	0.00	50.19	

11 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.72	6,398,250.00 *gal*
Application event totals		0.00	0.00	0.00	50.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.72	6,398,250.00 *gal*
Application event totals		0.00	0.00	0.00	50.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.46	6,364,575.00 *gal*
Application event totals		0.00	0.00	0.00	50.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.99	6,431,925.00 *gal*
Application event totals		0.00	0.00	0.00	50.99	

12-Brad - 11/15/2020: Alfalfa, hay

Field name: 12-Brad

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	299.31	127.73	265.00	0.00	990.00 *ton*
Application event totals		299.31	127.73	265.00	0.00	

12-Brad - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.92	6,061,500.00 *gal*
Application event totals		0.00	0.00	0.00	54.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.31	5,994,150.00 *gal*
Application event totals		0.00	0.00	0.00	54.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.61	6,027,825.00 *gal*
Application event totals		0.00	0.00	0.00	54.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.56	5,691,075.00 *gal*
Application event totals		0.00	0.00	0.00	51.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.17	5,758,425.00 *gal*
Application event totals		0.00	0.00	0.00	52.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.78	5,825,775.00 *gal*
Application event totals		0.00	0.00	0.00	52.78	

12-NE - 11/01/2022: Oats, silage-soft dough

12-NE - 11/01/2022: Oats, silage-soft dough

Field name: 12-NE

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	181.40	77.41	160.60	0.00	600.00 *ton*
Application event totals		181.40	77.41	160.60	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.21	5,320,650.00 *gal*
Application event totals		0.00	0.00	0.00	48.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.65	5,590,050.00 *gal*
Application event totals		0.00	0.00	0.00	50.65	

12-NE - 06/02/2023: Corn, silage

Field name: 12-NE

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	337.08	119.06	150.61	0.00	770.00 *ton*
Application event totals		337.08	119.06	150.61	0.00	

12-NE - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.94	4,849,200.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.02	4,748,175.00 *gal*
Application event totals		0.00	0.00	0.00	43.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.88	4,512,450.00 *gal*
Application event totals		0.00	0.00	0.00	40.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.94	4,849,200.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,714,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.41	4,680,825.00 *gal*
Application event totals		0.00	0.00	0.00	42.41	

12-NE - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.32	4,781,850.00 *gal*
Application event totals		0.00	0.00	0.00	43.32	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.94	4,849,200.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

12-NW - 11/01/2022: Oats, silage-soft dough

Field name: 12-NW

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	166.29	70.96	147.22	0.00	550.00 *ton*
Application event totals		166.29	70.96	147.22	0.00	

10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.51	5,354,325.00 *gal*
Application event totals		0.00	0.00	0.00	48.51	

04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.04	5,522,700.00 *gal*
Application event totals		0.00	0.00	0.00	50.04	

12-NW - 06/02/2023: Corn, silage

Field name: 12-NW

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	361.16	127.56	161.37	0.00	825.00 *ton*
Application event totals		361.16	127.56	161.37	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.85	4,950,225.00 *gal*
Application event totals		0.00	0.00	0.00	44.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.55	4,916,550.00 *gal*
Application event totals		0.00	0.00	0.00	44.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.55	4,916,550.00 *gal*
Application event totals		0.00	0.00	0.00	44.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.16	4,983,900.00 *gal*
Application event totals		0.00	0.00	0.00	45.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.24	4,882,875.00 *gal*
Application event totals		0.00	0.00	0.00	44.24	

12-NW - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.32	4,781,850.00 *gal*
Application event totals		0.00	0.00	0.00	43.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.94	4,849,200.00 *gal*
Application event totals		0.00	0.00	0.00	43.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.32	4,781,850.00 *gal*
Application event totals		0.00	0.00	0.00	43.32	

12-SE - 11/01/2022: Oats, silage-soft dough

Field name: 12-SE

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	151.17	64.51	133.84	0.00	500.00 *ton*
Application event totals		151.17	64.51	133.84	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.51	5,354,325.00 *gal*
Application event totals		0.00	0.00	0.00	48.51	

12-SE - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.43	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	49.43	

12-SE - 06/02/2023: Corn, silage

Field name: 12-SE

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	337.08	119.06	150.61	0.00	770.00 *ton*
Application event totals		337.08	119.06	150.61	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,714,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.80	4,613,475.00 *gal*
Application event totals		0.00	0.00	0.00	41.80	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,714,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

12-SE - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.41	4,680,825.00 *gal*
Application event totals		0.00	0.00	0.00	42.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.41	4,680,825.00 *gal*
Application event totals		0.00	0.00	0.00	42.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.80	4,613,475.00 *gal*
Application event totals		0.00	0.00	0.00	41.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.80	4,613,475.00 *gal*
Application event totals		0.00	0.00	0.00	41.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.41	4,680,825.00 *gal*
Application event totals		0.00	0.00	0.00	42.41	

12-SW - 11/01/2022: Oats, silage-soft dough

Field name: 12-SW

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

12-SW - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	176.36	75.26	156.14	0.00	750.00 *ton*
Application event totals		176.36	75.26	156.14	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.88	6,937,050.00 *gal*
Application event totals		0.00	0.00	0.00	48.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.94	6,802,350.00 *gal*
Application event totals		0.00	0.00	0.00	47.94	

12-SW - 06/02/2023: Corn, silage

Field name: 12-SW

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	337.08	119.06	150.61	0.00	990.00 *ton*
Application event totals		337.08	119.06	150.61	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.56	6,465,600.00 *gal*
Application event totals		0.00	0.00	0.00	45.56	

12-SW - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.51	6,600,300.00 *gal*
Application event totals		0.00	0.00	0.00	46.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.99	6,667,650.00 *gal*
Application event totals		0.00	0.00	0.00	46.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.27	6,566,625.00 *gal*
Application event totals		0.00	0.00	0.00	46.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.51	6,600,300.00 *gal*
Application event totals		0.00	0.00	0.00	46.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.04	6,532,950.00 *gal*
Application event totals		0.00	0.00	0.00	46.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.04	6,532,950.00 *gal*
Application event totals		0.00	0.00	0.00	46.04	

12-SW - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.80	6,499,275.00 *gal*
Application event totals		0.00	0.00	0.00	45.80	

13-E - 10/31/2022: Oats, silage-soft dough

Field name: 13-E

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	171.95	73.38	152.24	0.00	650.00 *ton*
Application event totals		171.95	73.38	152.24	0.00	

10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.19	5,825,775.00 *gal*
Application event totals		0.00	0.00	0.00	46.19	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.79	6,027,825.00 *gal*
Application event totals		0.00	0.00	0.00	47.79	

13-E - 06/01/2023: Corn, silage

Field name: 13-E

Crop: Corn, silage

Plant date: 06/01/2023

13-E - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	306.44	108.23	136.92	0.00	800.00 *ton*
Application event totals		306.44	108.23	136.92	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.05	5,556,375.00 *gal*
Application event totals		0.00	0.00	0.00	44.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	5,388,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	5,388,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	43.25	

13-E - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.45	5,354,325.00 *gal*
Application event totals		0.00	0.00	0.00	42.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.98	5,421,675.00 *gal*
Application event totals		0.00	0.00	0.00	42.98	

13-N - 10/31/2022: Oats, silage-soft dough

Field name: 13-N

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.73	67.74	140.53	0.00	600.00 *ton*
Application event totals		158.73	67.74	140.53	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.52	5,994,150.00 *gal*
Application event totals		0.00	0.00	0.00	47.52	

13-N - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.85	6,162,525.00 *gal*
Application event totals		0.00	0.00	0.00	48.85	

13-N - 06/01/2023: Corn, silage

Field name: 13-N

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	251.32	107.25	222.50	0.00	950.00 *ton*
Application event totals		251.32	107.25	222.50	0.00	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	43.25	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	5,522,700.00 *gal*
Application event totals		0.00	0.00	0.00	43.78	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

13-N - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	5,388,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	43.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	5,388,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

13-S - 10/31/2022: Oats, silage-soft dough

Field name: 13-S

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

13-S - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	185.18	79.03	163.95	0.00	700.00 *ton*
Application event totals		185.18	79.03	163.95	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.59	6,128,850.00 *gal*
Application event totals		0.00	0.00	0.00	48.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.12	6,196,200.00 *gal*
Application event totals		0.00	0.00	0.00	49.12	

13-S - 06/01/2023: Corn, silage

Field name: 13-S

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	325.59	115.00	145.48	0.00	850.00 *ton*
Application event totals		325.59	115.00	145.48	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	5,522,700.00 *gal*
Application event totals		0.00	0.00	0.00	43.78	

13-S - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.05	5,556,375.00 *gal*
Application event totals		0.00	0.00	0.00	44.05	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.05	5,556,375.00 *gal*
Application event totals		0.00	0.00	0.00	44.05	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	43.25	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	5,388,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,489,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.52	

13-S - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	43.25	

14 - 05/08/2022: Alfalfa, hay

Field name: 14

Crop: Alfalfa, hay

Plant date: 05/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	155.03	21.24	240.14	1,134.59	396,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.11	2,559,300.00 *gal*
Application event totals		155.03	21.24	240.14	1,188.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.39	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	53.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.11	2,559,300.00 *gal*
Application event totals		0.00	0.00	0.00	54.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.68	2,491,950.00 *gal*
Application event totals		0.00	0.00	0.00	52.68	

14 - 05/08/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.26	2,424,600.00 *gal*
Application event totals		0.00	0.00	0.00	51.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.39	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	53.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	193.79	26.55	300.17	1,418.23	495,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.26	2,424,600.00 *gal*
Application event totals		193.79	26.55	300.17	1,469.49	

16 - 10/31/2022: Oats, silage-soft dough

Field name: 16

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	169.31	72.25	149.90	0.00	160.00 *ton*
Application event totals		169.31	72.25	149.90	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.12	1,549,050.00 *gal*
Application event totals		0.00	0.00	0.00	49.12	

16 - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.39	1,683,750.00 *gal*
Application event totals		0.00	0.00	0.00	53.39	

16 - 06/01/2023: Corn, silage

Field name: 16

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	275.80	97.41	123.23	0.00	180.00 *ton*
Application event totals		275.80	97.41	123.23	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.51	1,245,975.00 *gal*
Application event totals		0.00	0.00	0.00	39.51	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	1,347,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.58	1,279,650.00 *gal*
Application event totals		0.00	0.00	0.00	40.58	

16 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.85	1,414,350.00 *gal*
Application event totals		0.00	0.00	0.00	44.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	1,347,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	1,347,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.65	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	41.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.92	1,448,025.00 *gal*
Application event totals		0.00	0.00	0.00	45.92	

17 - 05/08/2022: Alfalfa, hay

Field name: 17

Crop: Alfalfa, hay

Plant date: 05/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

17 - 05/08/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	284.89	121.58	252.23	0.00	350.00 *ton*
Application event totals		284.89	121.58	252.23	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.82	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	46.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.64	1,953,150.00 *gal*
Application event totals		0.00	0.00	0.00	47.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.00	1,885,800.00 *gal*
Application event totals		0.00	0.00	0.00	46.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.82	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	46.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.64	1,953,150.00 *gal*
Application event totals		0.00	0.00	0.00	47.64	

17 - 05/08/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.00	1,885,800.00 *gal*
Application event totals		0.00	0.00	0.00	46.00	

10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.36	1,818,450.00 *gal*
Application event totals		0.00	0.00	0.00	44.36	

2 - 11/02/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.89	11.91	125.32	845.90	1,314,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.14	6,196,200.00 *gal*
Application event totals		55.89	11.91	125.32	902.04	

04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	147.64	18.45	263.89	1,248.37	1,296,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	57.66	6,364,575.00 *gal*
Application event totals		147.64	18.45	263.89	1,306.04	

2 - 06/03/2023: Corn, silage

2 - 06/03/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.38	5,118,600.00 *gal*
Application event totals		0.00	0.00	0.00	46.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.85	4,950,225.00 *gal*
Application event totals		0.00	0.00	0.00	44.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.32	4,781,850.00 *gal*
Application event totals		0.00	0.00	0.00	43.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	128.16	16.01	229.07	1,083.66	1,125,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,714,500.00 *gal*
Application event totals		128.16	16.01	229.07	1,126.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	129.19	16.14	230.90	1,092.33	1,134,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.63	4,815,525.00 *gal*
Application event totals		129.19	16.14	230.90	1,135.96	

2 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.00	10.44	109.87	741.61	1,152,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,714,500.00 *gal*
Application event totals		49.00	10.44	109.87	784.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.32	4,781,850.00 *gal*
Application event totals		0.00	0.00	0.00	43.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.41	4,680,825.00 *gal*
Application event totals		0.00	0.00	0.00	42.41	

3 - 11/03/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.64	13.69	163.17	873.32	1,224,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	55.26	6,970,725.00 *gal*
Application event totals		93.64	13.69	163.17	928.58	

3 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	110.35	13.79	197.23	933.03	1,107,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.06	7,071,750.00 *gal*
Application event totals		110.35	13.79	197.23	989.09	

3 - 06/04/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	5,522,700.00 *gal*
Application event totals		0.00	0.00	0.00	43.78	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.32	5,590,050.00 *gal*
Application event totals		0.00	0.00	0.00	44.32	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	5,522,700.00 *gal*
Application event totals		0.00	0.00	0.00	43.78	

3 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	147.13	18.38	262.97	1,244.04	1,476,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		147.13	18.38	262.97	1,287.29	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	145.33	18.16	259.77	1,228.87	1,458,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	5,522,700.00 *gal*
Application event totals		145.33	18.16	259.77	1,272.65	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.60	11.84	124.67	841.55	1,494,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.25	5,455,350.00 *gal*
Application event totals		55.60	11.84	124.67	884.80	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.98	5,421,675.00 *gal*
Application event totals		0.00	0.00	0.00	42.98	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.58	5,623,725.00 *gal*
Application event totals		0.00	0.00	0.00	44.58	

4 - 11/03/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

4 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	82.62	12.08	143.98	770.58	810,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	61.22	5,792,100.00 *gal*
Application event totals		82.62	12.08	143.98	831.80	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.48	12.55	179.59	849.59	756,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	58.73	5,556,375.00 *gal*
Application event totals		100.48	12.55	179.59	908.32	

4 - 06/04/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.99	4,445,100.00 *gal*
Application event totals		0.00	0.00	0.00	46.99	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.92	4,344,075.00 *gal*
Application event totals		0.00	0.00	0.00	45.92	

4 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.34	4,478,775.00 *gal*
Application event totals		0.00	0.00	0.00	47.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	129.19	16.14	230.90	1,092.33	972,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	4,142,025.00 *gal*
Application event totals		129.19	16.14	230.90	1,136.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.01	15.24	218.07	1,031.64	918,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.85	4,243,050.00 *gal*
Application event totals		122.01	15.24	218.07	1,076.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.22	9.42	99.14	669.19	891,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.14	4,175,700.00 *gal*
Application event totals		44.22	9.42	99.14	713.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.85	4,243,050.00 *gal*
Application event totals		0.00	0.00	0.00	44.85	

4 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.27	4,377,750.00 *gal*
Application event totals		0.00	0.00	0.00	46.27	

5 - 11/03/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.13	12.30	146.59	784.59	1,512,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.76	8,284,050.00 *gal*
Application event totals		84.13	12.30	146.59	832.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.39	13.04	186.59	882.69	1,440,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.51	8,587,125.00 *gal*
Application event totals		104.39	13.04	186.59	932.20	

5 - 06/04/2023: Corn, silage

Field name: 5

Crop: Corn, silage Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.07	7,644,225.00 *gal*
Application event totals		0.00	0.00	0.00	44.07	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.69	7,576,875.00 *gal*
Application event totals		0.00	0.00	0.00	43.69	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.49	7,543,200.00 *gal*
Application event totals		0.00	0.00	0.00	43.49	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	126.58	15.82	226.24	1,070.26	1,746,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.07	7,644,225.00 *gal*
Application event totals		126.58	15.82	226.24	1,114.33	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.01	15.24	218.07	1,031.64	1,683,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.10	7,475,850.00 *gal*
Application event totals		122.01	15.24	218.07	1,074.75	

5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.29	9.86	103.78	700.52	1,710,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.10	7,475,850.00 *gal*
Application event totals		46.29	9.86	103.78	743.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.69	7,576,875.00 *gal*
Application event totals		0.00	0.00	0.00	43.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.07	7,644,225.00 *gal*
Application event totals		0.00	0.00	0.00	44.07	

6 - 04/30/2022: Alfalfa, hay

Field name: 6

Crop: Alfalfa, hay

Plant date: 04/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	213.17	29.20	330.19	1,560.06	1,089,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.14	4,175,700.00 *gal*
Application event totals		213.17	29.20	330.19	1,604.19	

6 - 04/30/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.43	4,108,350.00 *gal*
Application event totals		0.00	0.00	0.00	43.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,041,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.78	4,142,025.00 *gal*
Application event totals		0.00	0.00	0.00	43.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.07	4,074,675.00 *gal*
Application event totals		0.00	0.00	0.00	43.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	4,041,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.92	10.85	114.16	770.58	1,026,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.43	4,108,350.00 *gal*
Application event totals		50.92	10.85	114.16	814.00	

7 - 11/03/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.06	12.37	130.18	878.73	819,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.82	3,232,800.00 *gal*
Application event totals		58.06	12.37	130.18	927.55	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	118.03	17.25	205.68	1,100.82	810,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.34	3,333,825.00 *gal*
Application event totals		118.03	17.25	205.68	1,151.17	

7 - 06/04/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.75	2,963,400.00 *gal*
Application event totals		0.00	0.00	0.00	44.75	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	2,828,700.00 *gal*
Application event totals		0.00	0.00	0.00	42.71	

7 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.22	2,862,375.00 *gal*
Application event totals		0.00	0.00	0.00	43.22	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	128.16	16.01	229.07	1,083.66	675,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.24	2,929,725.00 *gal*
Application event totals		128.16	16.01	229.07	1,127.90	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	123.03	15.37	219.91	1,040.31	648,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.73	2,896,050.00 *gal*
Application event totals		123.03	15.37	219.91	1,084.04	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.49	10.33	108.72	733.88	684,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.71	2,828,700.00 *gal*
Application event totals		48.49	10.33	108.72	776.60	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.73	2,896,050.00 *gal*
Application event totals		0.00	0.00	0.00	43.73	

7 - 06/04/2023: Corn, silage

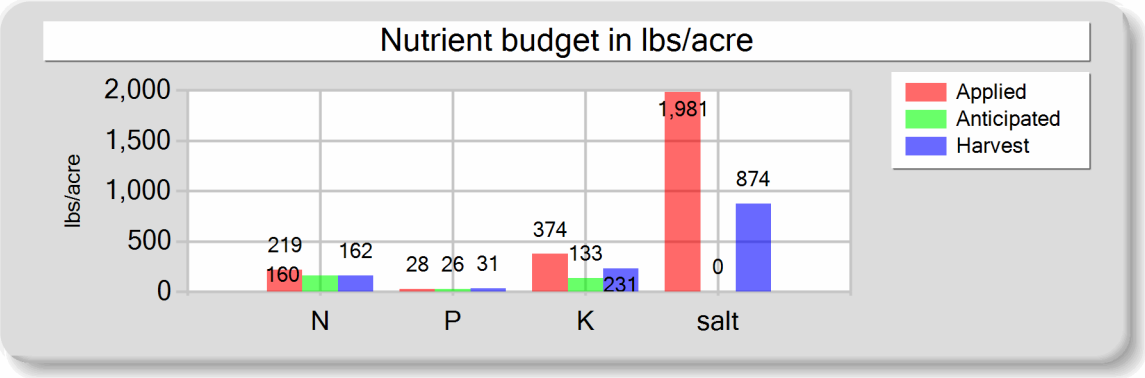
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.75	2,963,400.00 *gal*
Application event totals		0.00	0.00	0.00	44.75	

B. NUTRIENT BUDGET

1 - 11/02/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.79	28.43	374.38	1,871.62
Fresh water	0.00	0.00	0.00	109.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.79	28.43	374.38	1,981.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	161.60	30.57	231.48	873.53
Nutrient balance	57.18	-2.14	142.89	1,107.93
Applied to removed ratio	1.35	0.93	1.62	2.27

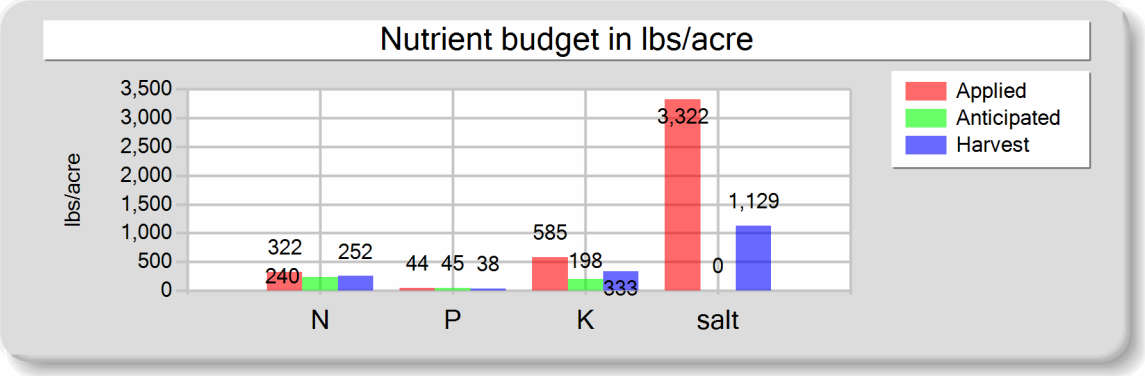
Fresh water applied	
12,123,000.00	*gallons*
446.45	*acre-inches*
12.76	*inches/acre*

Process wastewater applied	
2,106,000.00	*gallons*
77.56	*acre-inches*
2.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/03/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	314.55	43.62	584.50	2,986.95
Fresh water	0.00	0.00	0.00	335.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	321.55	43.62	584.50	3,322.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.07	38.32	332.74	1,129.29
Nutrient balance	69.48	5.30	251.77	2,192.97
Applied to removed ratio	1.28	1.14	1.76	2.94

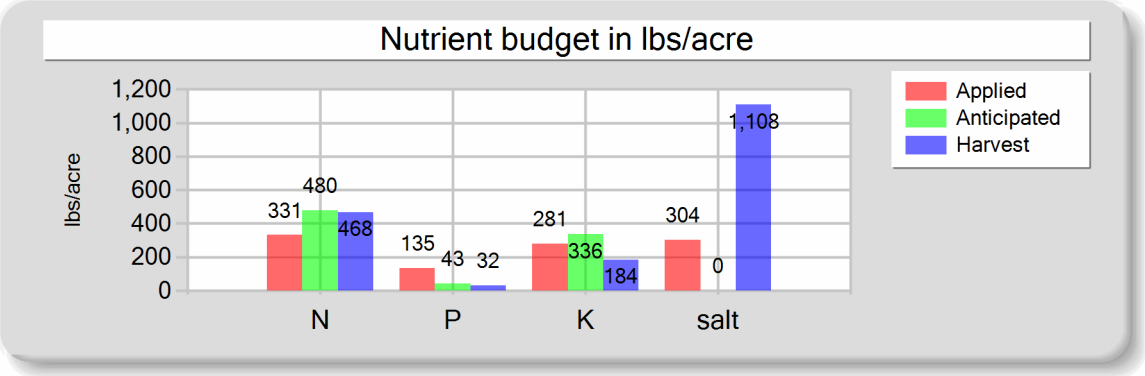
Fresh water applied	
37,008,825.00	*gallons*
1,362.91	*acre-inches*
38.94	*inches/acre*

Process wastewater applied	
3,483,000.00	*gallons*
128.27	*acre-inches*
3.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 11/15/2020: Alfalfa, hay

Field name: 11 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	317.45	135.47	281.06	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	303.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	331.45	135.47	281.06	303.54
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	467.96	32.48	184.13	1,108.24
Nutrient balance	-136.50	103.00	96.93	-804.70
Applied to removed ratio	0.71	4.17	1.53	0.27

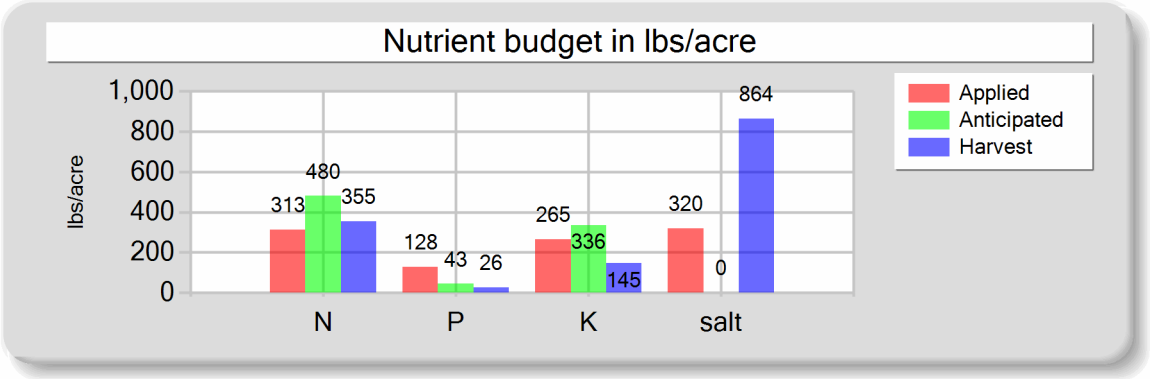
Fresh water applied	
38,288,475.00	*gallons*
1,410.03	*acre-inches*
35.25	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

12-Brad - 11/15/2020: Alfalfa, hay

Field name: 12-Brad Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	299.31	127.73	265.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	320.36
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	313.31	127.73	265.00	320.36
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	355.38	26.32	145.25	863.87
Nutrient balance	-42.06	101.41	119.74	-543.51
Applied to removed ratio	0.88	4.85	1.82	0.37

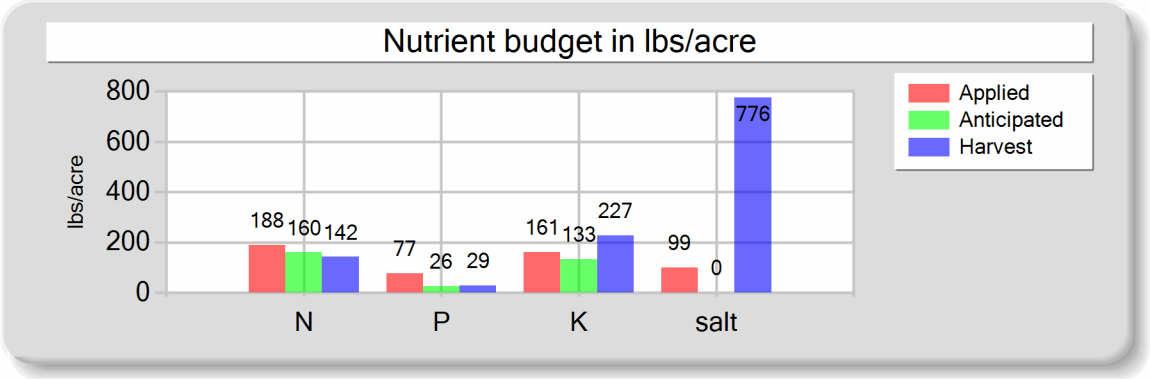
Fresh water applied	
35,358,750.00	*gallons*
1,302.14	*acre-inches*
37.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

12-NE - 11/01/2022: Oats, silage-soft dough

Field name: 12-NE Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	181.40	77.41	160.60	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	98.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.40	77.41	160.60	98.85
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.69	28.64	226.86	776.29
Nutrient balance	46.71	48.77	-66.25	-677.44
Applied to removed ratio	1.33	2.70	0.71	0.13

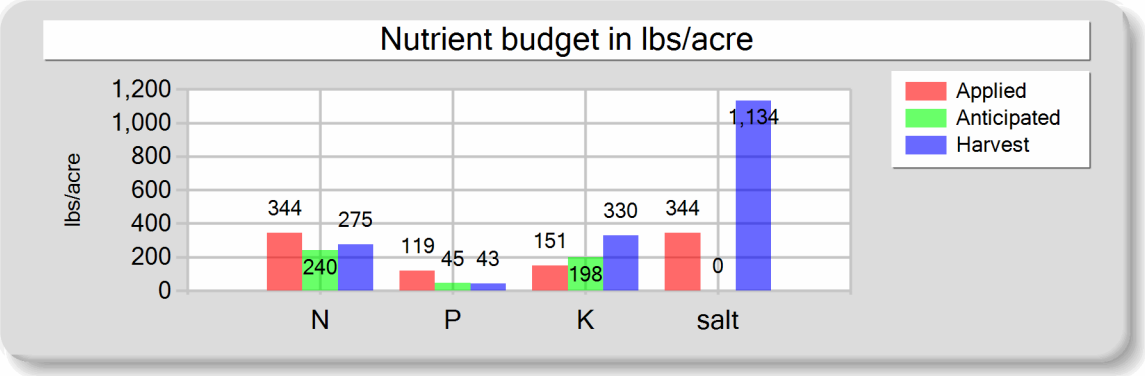
Fresh water applied	
10,910,700.00	*gallons*
401.80	*acre-inches*
11.48	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-NE - 06/02/2023: Corn, silage

Field name: 12-NE | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	337.08	119.06	150.61	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	344.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.08	119.06	150.61	344.16
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	275.42	42.53	330.10	1,134.07
Nutrient balance	68.67	76.53	-179.48	-789.91
Applied to removed ratio	1.25	2.80	0.46	0.30

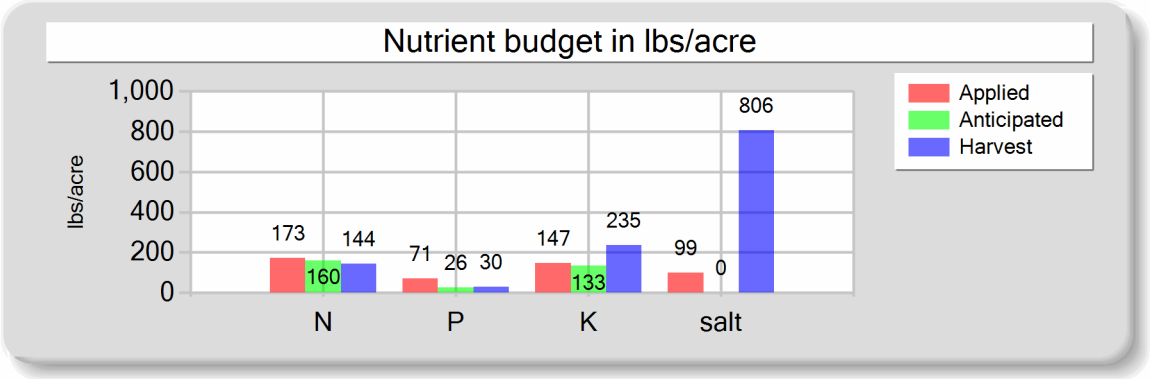
Fresh water applied	
37,985,400.00	*gallons*
1,398.87	*acre-inches*
39.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-NW - 11/01/2022: Oats, silage-soft dough

Field name: 12-NW Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.29	70.96	147.22	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	98.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.29	70.96	147.22	98.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.91	29.60	235.19	805.89
Nutrient balance	29.38	41.36	-87.97	-707.34
Applied to removed ratio	1.20	2.40	0.63	0.12

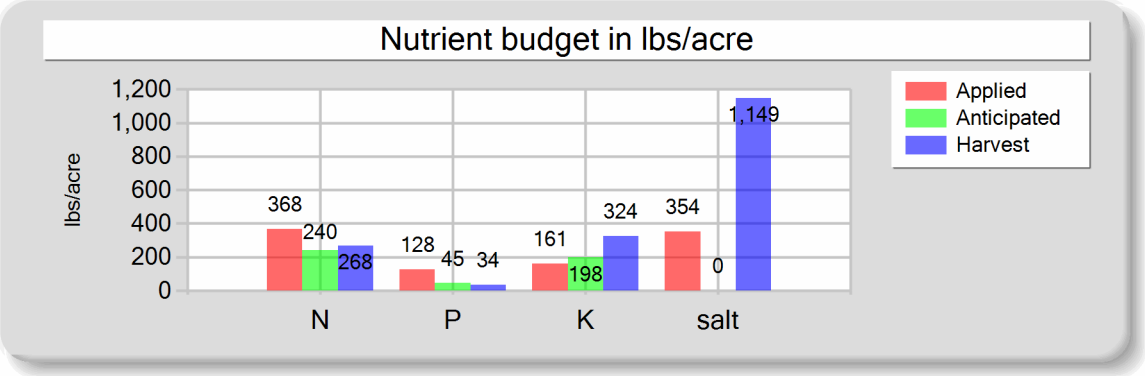
Fresh water applied	
10,877,025.00	*gallons*
400.56	*acre-inches*
11.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-NW - 06/02/2023: Corn, silage

Field name: 12-NW Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	361.16	127.56	161.37	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	353.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	368.16	127.56	161.37	353.92
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.07	34.26	324.50	1,148.85
Nutrient balance	100.10	93.30	-163.13	-794.93
Applied to removed ratio	1.37	3.72	0.50	0.31

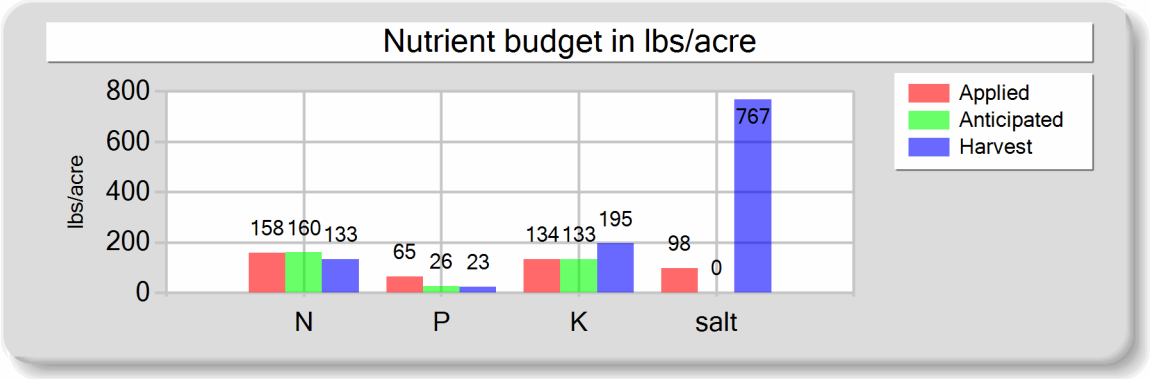
Fresh water applied	
39,063,000.00	*gallons*
1,438.56	*acre-inches*
41.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-SE - 11/01/2022: Oats, silage-soft dough

Field name: 12-SE Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	151.17	64.51	133.84	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	97.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.17	64.51	133.84	97.94
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	133.46	23.43	195.22	766.69
Nutrient balance	24.71	41.08	-61.38	-668.75
Applied to removed ratio	1.19	2.75	0.69	0.13

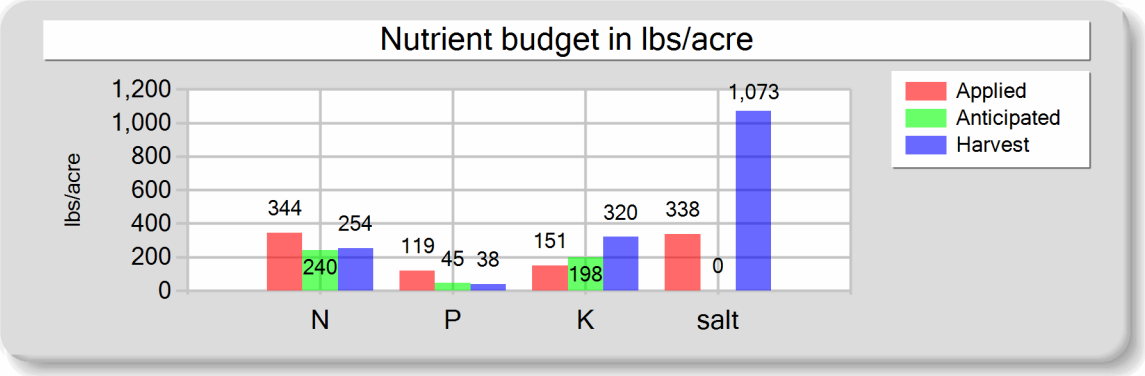
Fresh water applied	
10,809,675.00	*gallons*
398.08	*acre-inches*
11.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-SE - 06/02/2023: Corn, silage

Field name: 12-SE Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	337.08	119.06	150.61	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	338.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.08	119.06	150.61	338.06
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	254.25	37.74	319.79	1,072.60
Nutrient balance	89.84	81.32	-169.18	-734.55
Applied to removed ratio	1.35	3.15	0.47	0.32

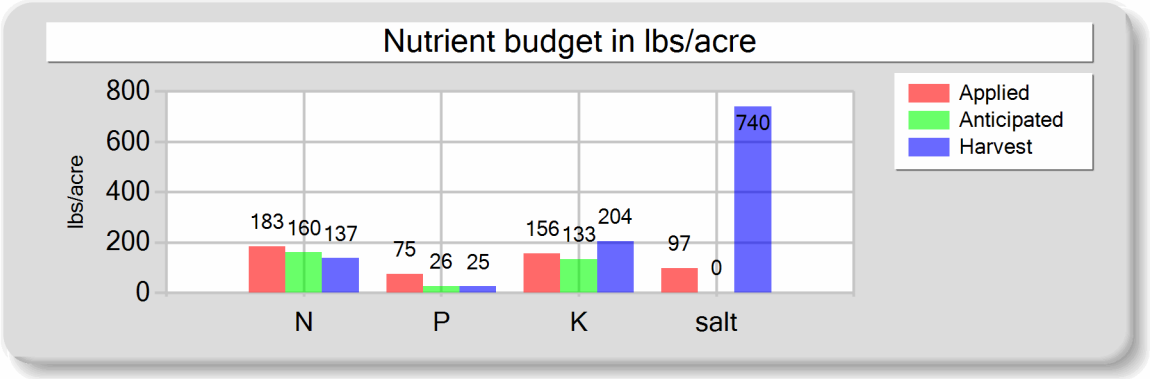
Fresh water applied	
37,311,900.00	*gallons*
1,374.07	*acre-inches*
39.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-SW - 11/01/2022: Oats, silage-soft dough

Field name: 12-SW | Crop: Oats, silage-soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.36	75.26	156.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	96.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.36	75.26	156.14	96.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.94	24.90	204.02	740.01
Nutrient balance	46.43	50.37	-47.88	-643.19
Applied to removed ratio	1.34	3.02	0.77	0.13

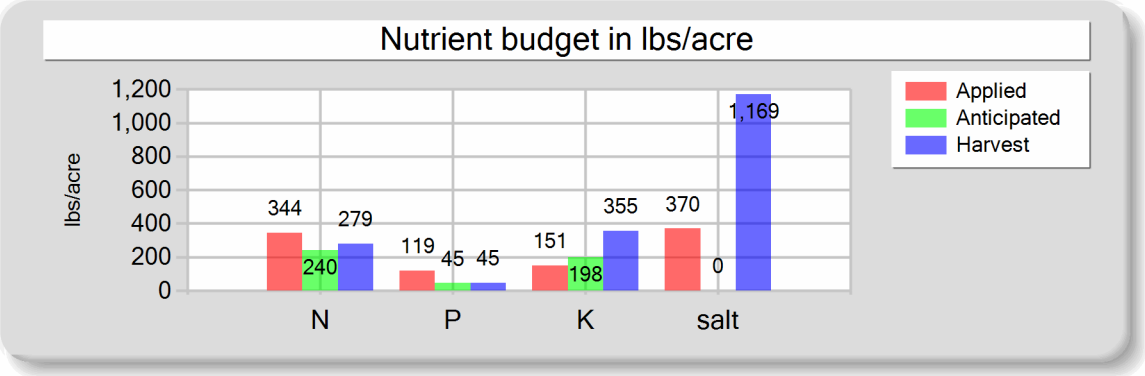
Fresh water applied	
13,739,400.00	*gallons*
505.98	*acre-inches*
11.24	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12-SW - 06/02/2023: Corn, silage

Field name: 12-SW | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	337.08	119.06	150.61	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	369.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.08	119.06	150.61	369.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	279.04	45.14	354.95	1,169.50
Nutrient balance	65.05	73.92	-204.34	-799.78
Applied to removed ratio	1.23	2.64	0.42	0.32

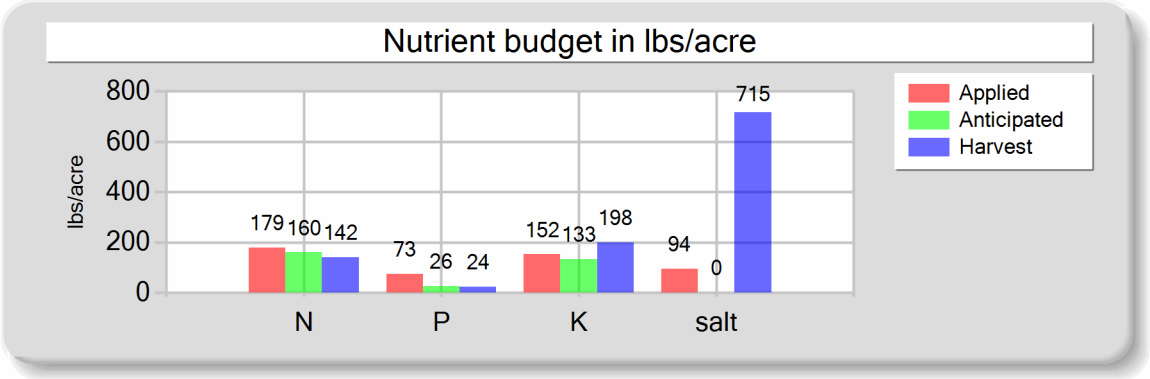
Fresh water applied	
52,465,650.00	*gallons*
1,932.13	*acre-inches*
42.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13-E - 10/31/2022: Oats, silage-soft dough

Field name: 13-E | Crop: Oats, silage-soft dough | Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	171.95	73.38	152.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	93.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.95	73.38	152.24	93.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.59	24.34	198.38	715.34
Nutrient balance	37.36	49.05	-46.14	-621.37
Applied to removed ratio	1.26	3.02	0.77	0.13

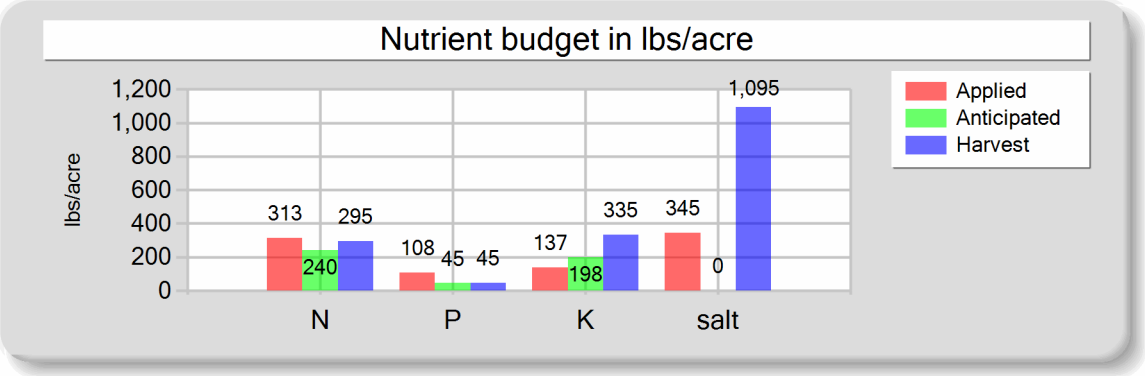
Fresh water applied	
11,853,600.00	*gallons*
436.53	*acre-inches*
10.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13-E - 06/01/2023: Corn, silage

Field name: 13-E Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	306.44	108.23	136.92	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	345.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.44	108.23	136.92	345.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	295.39	45.44	334.64	1,094.80
Nutrient balance	18.05	62.79	-197.72	-749.61
Applied to removed ratio	1.06	2.38	0.41	0.32

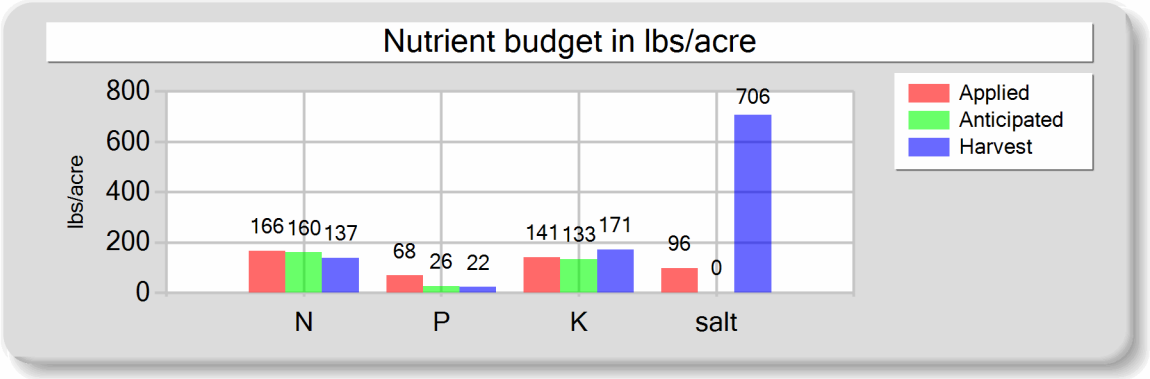
Fresh water applied	
43,541,775.00	*gallons*
1,603.50	*acre-inches*
40.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13-N - 10/31/2022: Oats, silage-soft dough

Field name: 13-N Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.73	67.74	140.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	96.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.73	67.74	140.53	96.38
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	137.30	21.99	171.29	706.47
Nutrient balance	28.43	45.74	-30.76	-610.10
Applied to removed ratio	1.21	3.08	0.82	0.14

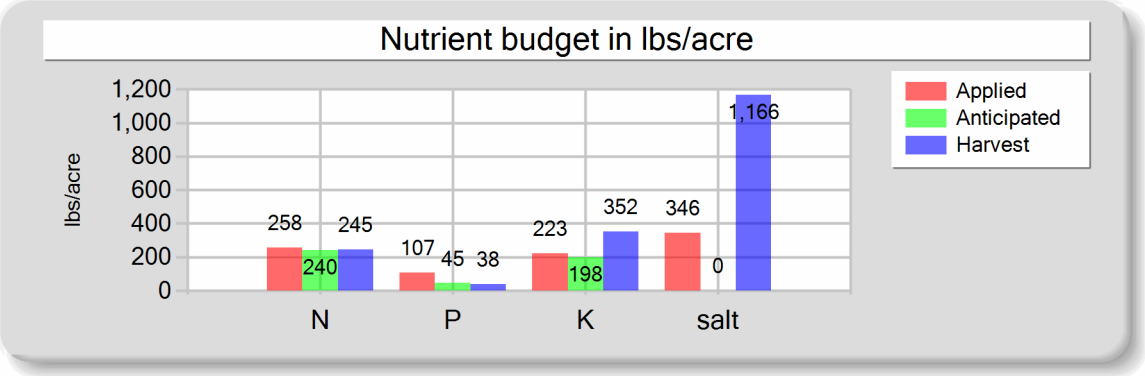
Fresh water applied	
12,156,675.00	*gallons*
447.69	*acre-inches*
11.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13-N - 06/01/2023: Corn, silage

Field name: 13-N Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	251.32	107.25	222.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	346.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.32	107.25	222.50	346.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	245.36	38.21	351.95	1,166.46
Nutrient balance	12.96	69.04	-129.44	-820.20
Applied to removed ratio	1.05	2.81	0.63	0.30

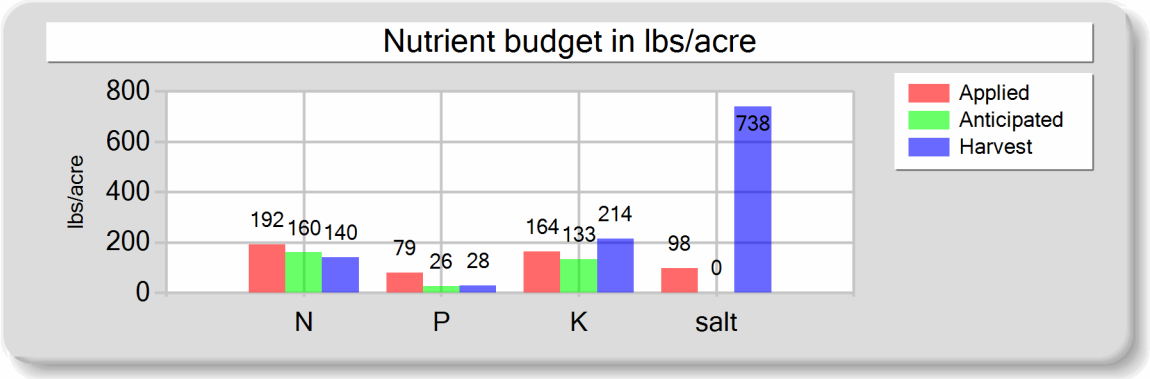
Fresh water applied	
43,676,475.00	*gallons*
1,608.46	*acre-inches*
40.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13-S - 10/31/2022: Oats, silage-soft dough

Field name: 13-S | Crop: Oats, silage-soft dough | Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	185.18	79.03	163.95	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	97.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.18	79.03	163.95	97.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	140.07	28.17	213.92	738.43
Nutrient balance	52.11	50.86	-49.97	-640.72
Applied to removed ratio	1.37	2.81	0.77	0.13

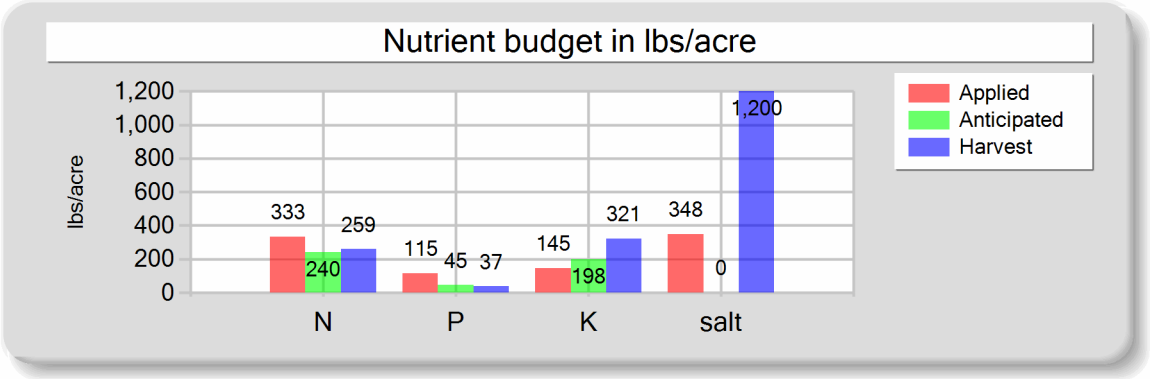
Fresh water applied	
12,325,050.00	*gallons*
453.89	*acre-inches*
11.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13-S - 06/01/2023: Corn, silage

Field name: 13-S | Crop: Corn, silage | Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	325.59	115.00	145.48	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	348.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	332.59	115.00	145.48	348.12
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	258.58	37.23	320.63	1,199.79
Nutrient balance	74.02	77.76	-175.16	-851.67
Applied to removed ratio	1.29	3.09	0.45	0.29

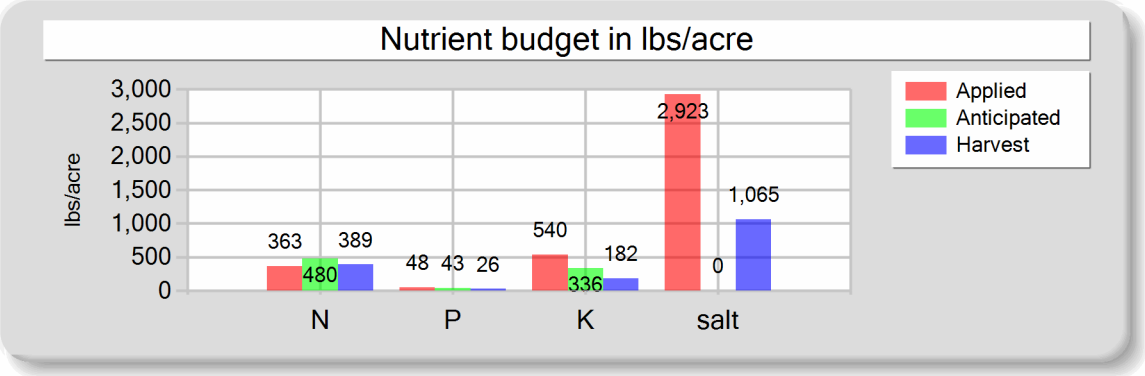
Fresh water applied	
43,912,200.00	*gallons*
1,617.14	*acre-inches*
40.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 05/08/2022: Alfalfa, hay

Field name: 14 Crop: Alfalfa, hay Plant date: 05/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	348.82	47.78	540.31	2,552.82
Fresh water	0.00	0.00	0.00	370.19
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	362.82	47.78	540.31	2,923.01
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	389.16	26.10	182.15	1,065.50
Nutrient balance	-26.34	21.69	358.16	1,857.51
Applied to removed ratio	0.93	1.83	2.97	2.74

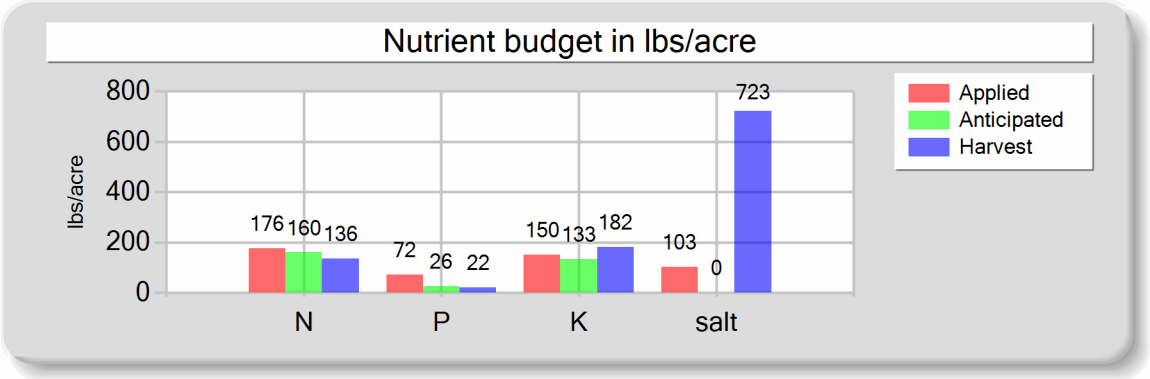
Fresh water applied	
17,511,000.00	*gallons*
644.87	*acre-inches*
42.99	*inches/acre*

Process wastewater applied	
891,000.00	*gallons*
32.81	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
6	*harvests*

16 - 10/31/2022: Oats, silage-soft dough

Field name: 16 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.31	72.25	149.90	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	102.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.31	72.25	149.90	102.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.03	21.76	181.89	722.89
Nutrient balance	40.28	50.49	-31.99	-620.38
Applied to removed ratio	1.30	3.32	0.82	0.14

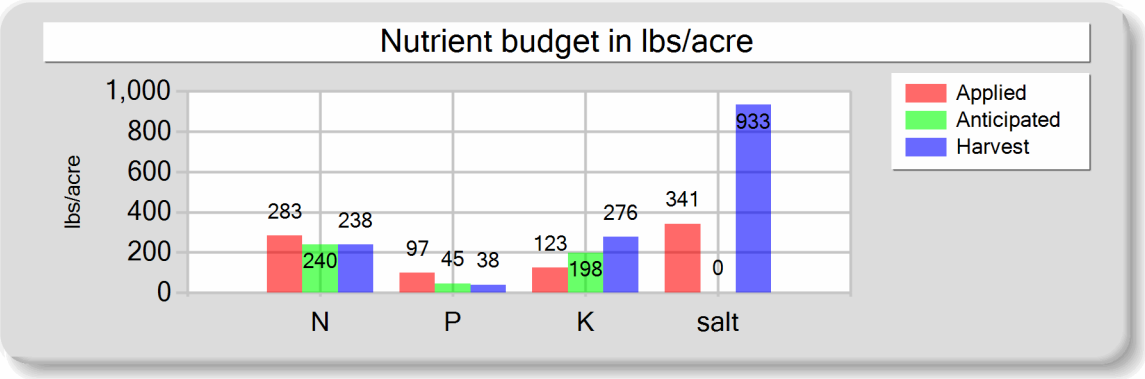
Fresh water applied	
3,232,800.00	*gallons*
119.05	*acre-inches*
11.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

16 - 06/01/2023: Corn, silage

Field name: 16 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	275.80	97.41	123.23	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	340.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.80	97.41	123.23	340.65
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	238.27	37.73	275.99	933.21
Nutrient balance	44.53	59.68	-152.76	-592.56
Applied to removed ratio	1.19	2.58	0.45	0.37

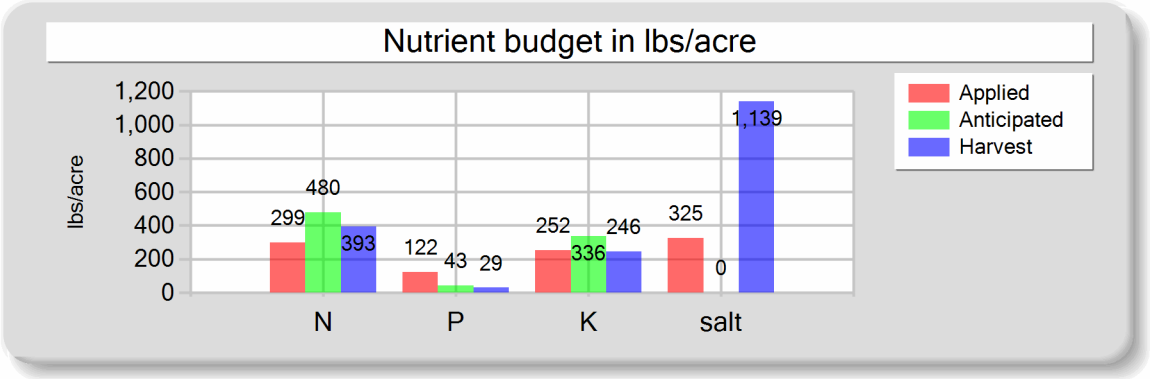
Fresh water applied	
10,742,325.00	*gallons*
395.60	*acre-inches*
39.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

17 - 05/08/2022: Alfalfa, hay

Field name: 17 Crop: Alfalfa, hay Plant date: 05/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.89	121.58	252.23	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	325.29
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	298.89	121.58	252.23	325.29
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	392.61	29.01	246.24	1,138.73
Nutrient balance	-93.72	92.57	6.00	-813.45
Applied to removed ratio	0.76	4.19	1.02	0.29

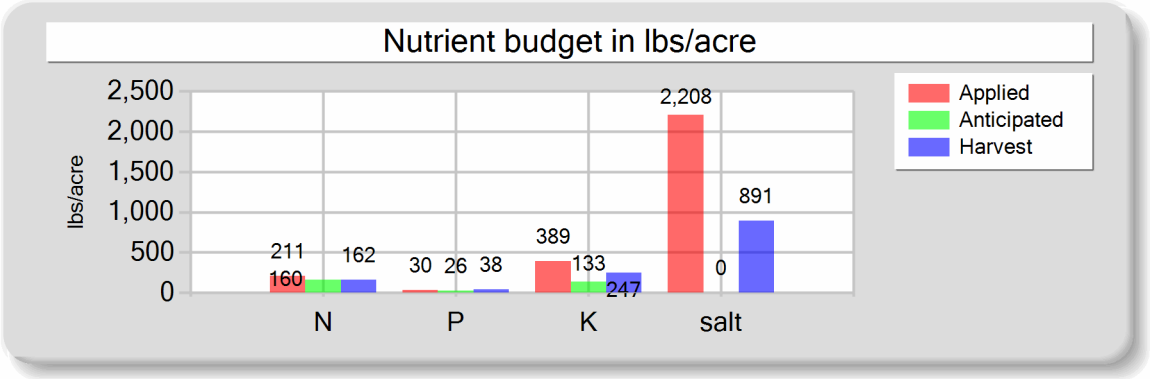
Fresh water applied
13,335,300.00 *gallons*
491.09 *acre-inches*
37.78 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
6 *harvests*

2 - 11/02/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.53	30.35	389.21	2,094.27
Fresh water	0.00	0.00	0.00	113.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.53	30.35	389.21	2,208.07
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.07	38.09	247.16	891.38
Nutrient balance	48.46	-7.73	142.05	1,316.69
Applied to removed ratio	1.30	0.80	1.57	2.48

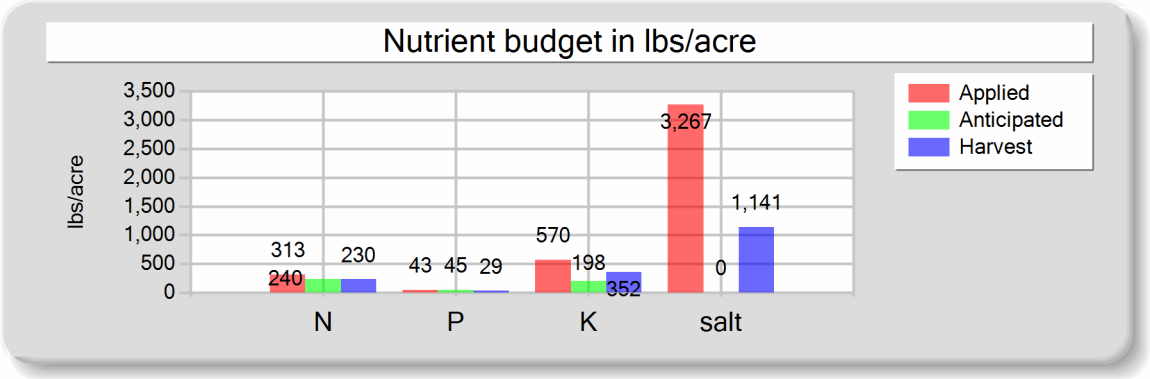
Fresh water applied	
12,560,775.00	*gallons*
462.57	*acre-inches*
13.22	*inches/acre*

Process wastewater applied	
2,610,000.00	*gallons*
96.12	*acre-inches*
2.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/03/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	306.35	42.59	569.84	2,917.59
Fresh water	0.00	0.00	0.00	349.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.35	42.59	569.84	3,266.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	230.18	29.01	352.04	1,141.23
Nutrient balance	83.17	13.58	217.80	2,125.71
Applied to removed ratio	1.36	1.47	1.62	2.86

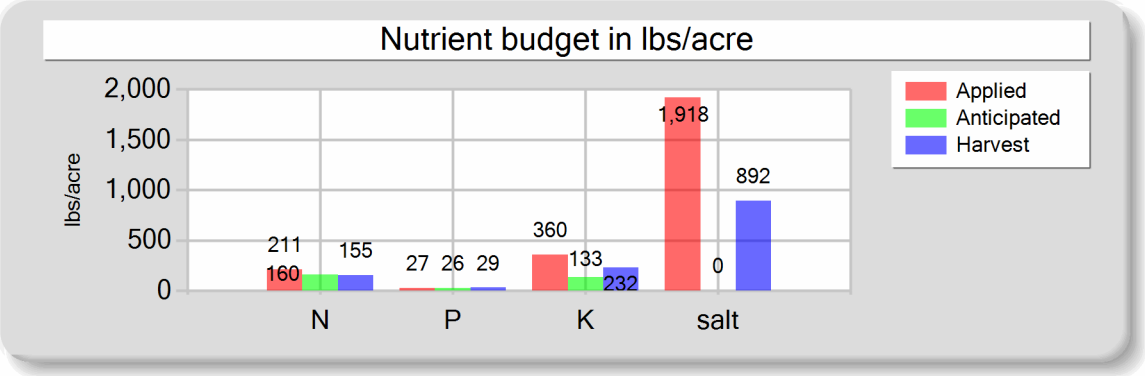
Fresh water applied	
38,557,875.00	*gallons*
1,419.96	*acre-inches*
40.57	*inches/acre*

Process wastewater applied	
3,411,000.00	*gallons*
125.62	*acre-inches*
3.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/03/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.99	27.47	360.40	1,806.35
Fresh water	0.00	0.00	0.00	111.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.99	27.47	360.40	1,917.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.73	29.13	232.10	891.99
Nutrient balance	56.25	-1.65	128.30	1,025.68
Applied to removed ratio	1.36	0.94	1.55	2.15

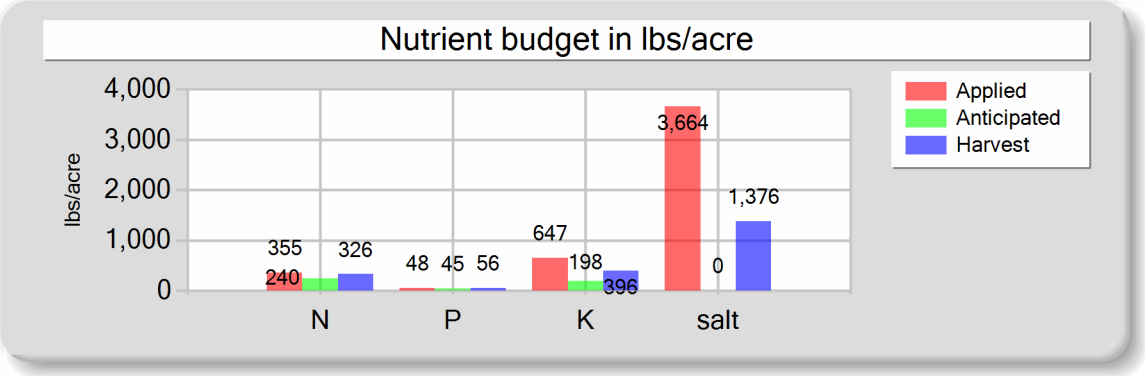
Fresh water applied	
14,042,475.00	*gallons*
517.14	*acre-inches*
12.93	*inches/acre*

Process wastewater applied	
2,331,000.00	*gallons*
85.84	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/04/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	348.07	48.39	647.41	3,314.46
Fresh water	0.00	0.00	0.00	349.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	355.07	48.39	647.41	3,664.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	326.48	55.97	396.44	1,375.88
Nutrient balance	28.59	-7.58	250.97	2,288.31
Applied to removed ratio	1.09	0.86	1.63	2.66

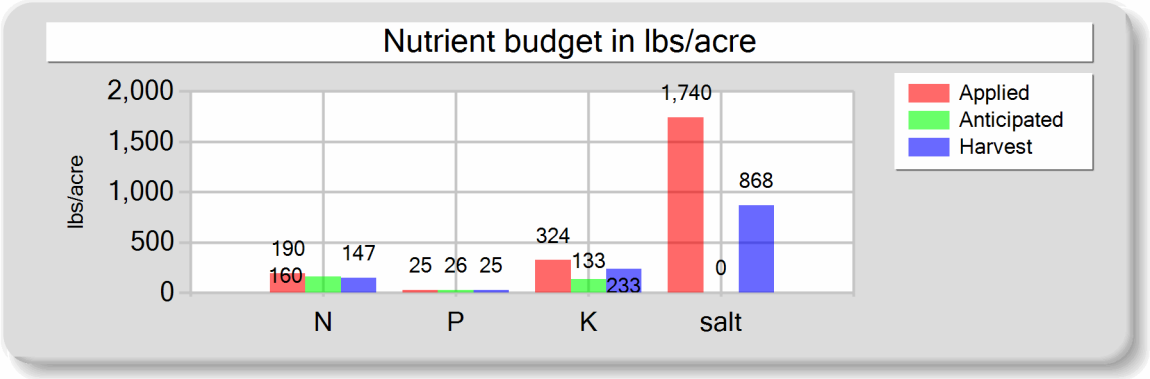
Fresh water applied	
44,114,250.00	*gallons*
1,624.58	*acre-inches*
40.61	*inches/acre*

Process wastewater applied	
4,428,000.00	*gallons*
163.07	*acre-inches*
4.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/03/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.10	24.63	323.57	1,620.17
Fresh water	0.00	0.00	0.00	119.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.10	24.63	323.57	1,740.12
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.64	25.16	233.41	867.70
Nutrient balance	43.46	-0.53	90.16	872.43
Applied to removed ratio	1.30	0.98	1.39	2.01

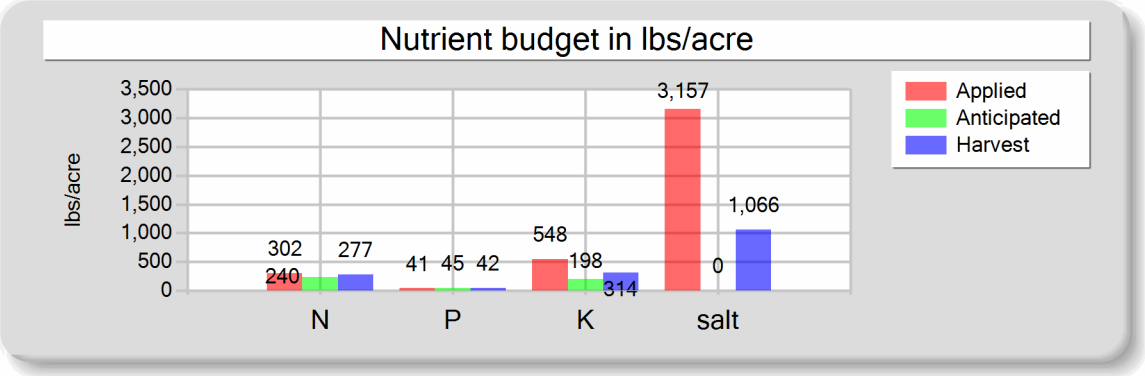
Fresh water applied	
11,348,475.00	*gallons*
417.93	*acre-inches*
13.93	*inches/acre*

Process wastewater applied	
1,566,000.00	*gallons*
57.67	*acre-inches*
1.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/04/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.41	40.80	548.12	2,793.15
Fresh water	0.00	0.00	0.00	364.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.41	40.80	548.12	3,157.30
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	277.48	42.22	313.67	1,065.67
Nutrient balance	24.94	-1.42	234.45	2,091.63
Applied to removed ratio	1.09	0.97	1.75	2.96

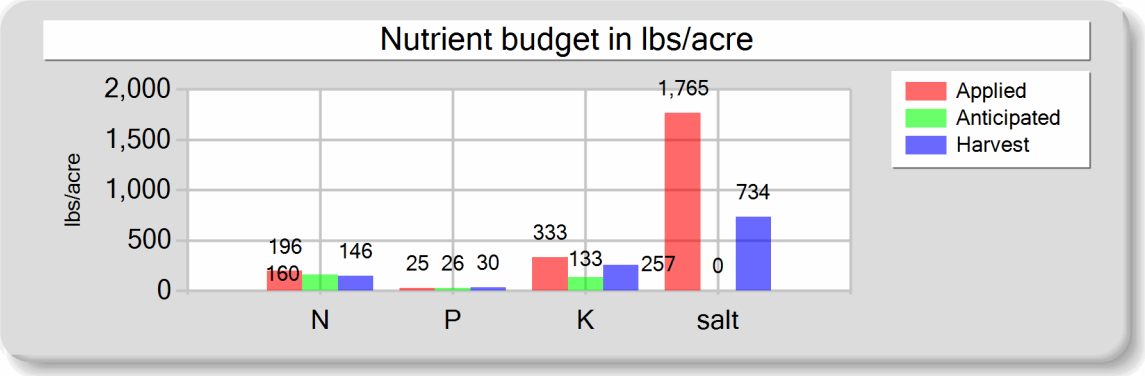
Fresh water applied	
34,449,525.00	*gallons*
1,268.66	*acre-inches*
42.29	*inches/acre*

Process wastewater applied	
2,781,000.00	*gallons*
102.41	*acre-inches*
3.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/03/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	188.52	25.34	333.18	1,667.28
Fresh water	0.00	0.00	0.00	97.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.52	25.34	333.18	1,764.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.11	30.41	256.62	734.25
Nutrient balance	49.41	-5.07	76.57	1,030.30
Applied to removed ratio	1.34	0.83	1.30	2.40

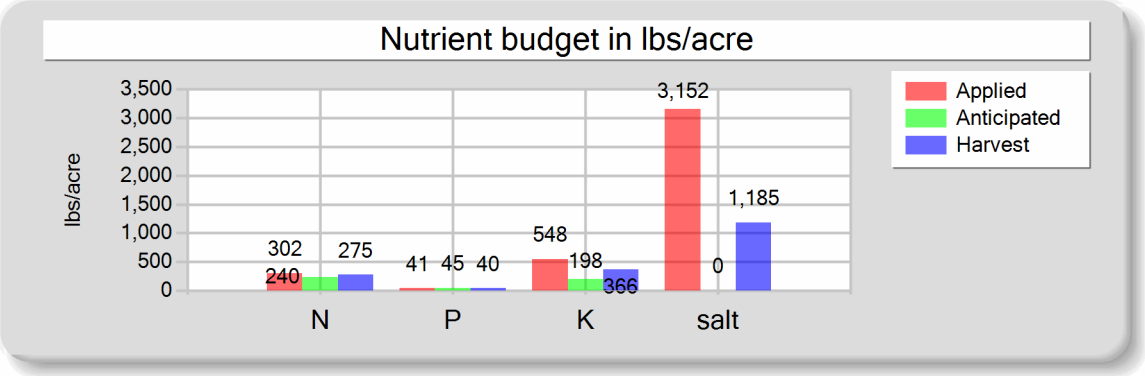
Fresh water applied	
16,871,175.00	*gallons*
621.31	*acre-inches*
11.30	*inches/acre*

Process wastewater applied	
2,952,000.00	*gallons*
108.71	*acre-inches*
1.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/04/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.87	40.92	548.09	2,802.43
Fresh water	0.00	0.00	0.00	349.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.87	40.92	548.09	3,151.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	275.07	40.20	366.06	1,184.93
Nutrient balance	26.80	0.72	182.04	1,966.79
Applied to removed ratio	1.10	1.02	1.50	2.66

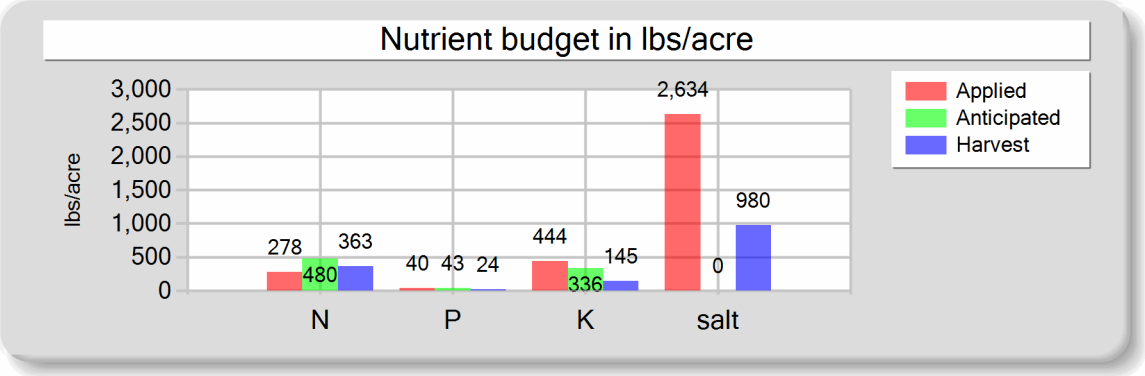
Fresh water applied	
60,581,325.00	*gallons*
2,231.00	*acre-inches*
40.56	*inches/acre*

Process wastewater applied	
5,139,000.00	*gallons*
189.25	*acre-inches*
3.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 04/30/2022: Alfalfa, hay

Field name: 6 Crop: Alfalfa, hay Plant date: 04/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	264.08	40.05	444.35	2,330.63
Fresh water	0.00	0.00	0.00	303.27
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	278.08	40.05	444.35	2,633.91
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	363.40	24.32	144.87	980.48
Nutrient balance	-85.32	15.72	299.47	1,653.43
Applied to removed ratio	0.77	1.65	3.07	2.69

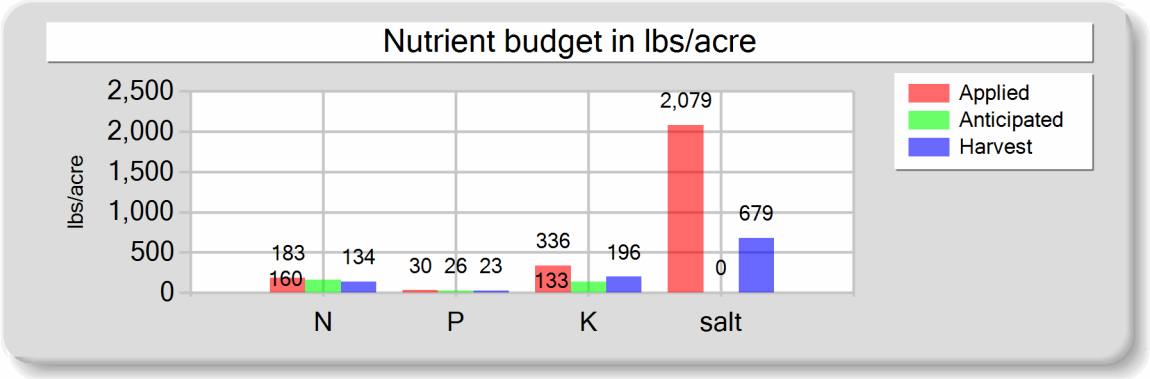
Fresh water applied	
28,691,100.00	*gallons*
1,056.60	*acre-inches*
35.22	*inches/acre*

Process wastewater applied	
2,115,000.00	*gallons*
77.89	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
6	*harvests*

7 - 11/03/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	176.09	29.62	335.86	1,979.55
Fresh water	0.00	0.00	0.00	99.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.09	29.62	335.86	2,078.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.40	23.32	196.12	678.88
Nutrient balance	48.69	6.30	139.74	1,399.83
Applied to removed ratio	1.36	1.27	1.71	3.06

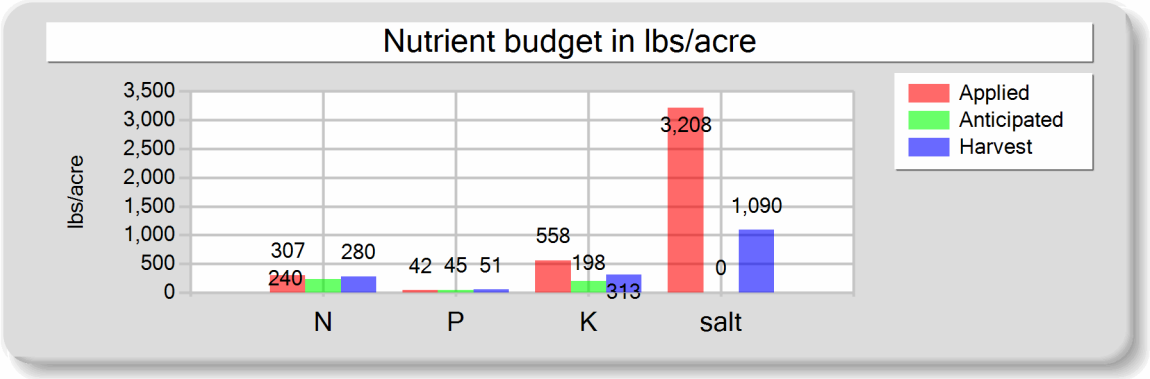
Fresh water applied	
6,566,625.00	*gallons*
241.83	*acre-inches*
11.52	*inches/acre*

Process wastewater applied	
1,629,000.00	*gallons*
59.99	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/04/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	299.69	41.72	557.70	2,857.85
Fresh water	0.00	0.00	0.00	349.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.69	41.72	557.70	3,207.71
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	279.67	51.41	312.58	1,089.90
Nutrient balance	27.01	-9.70	245.13	2,117.80
Applied to removed ratio	1.10	0.81	1.78	2.94

Fresh water applied	
23,168,400.00	*gallons*
853.21	*acre-inches*
40.63	*inches/acre*

Process wastewater applied	
2,007,000.00	*gallons*
73.91	*acre-inches*
3.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	8,300.00	10,500.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	6,700.00	13,900.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	702.00	205.00	0.00	1.70	96.40	1,090.00								10,400.00	5,150
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	477.00	375.00	0.00	0.80	59.70	854.00								8,850.00	4,040
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	178.00	153.00	0.00	0.40	38.00	400.00								5,220.00	2,700
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	366.00	280.00	0.00	0.70	53.60	639.00								6,530.00	3,420
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well #1

Domestic Well - Well #1

Well #1

Sample description: Well #1

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			38.00								1,620.00	
DL			0.10								10.00	

Domestic Well - Well #3

Well #3

Sample description: Well #3

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			69.80								1,810.00	
DL			0.10								10.00	

Domestic Well - Well #4

Well #4

Sample description: Well #4

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			71.70								1,860.00	
DL			0.10								10.00	

Domestic Well - Well #5

Domestic Well - Well #5

Well #5

Sample description: Well #5

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			60.50								1,830.00	
DL			0.10								10.00	

Domestic Well - Well #7

Well #7

Sample description: Well #7

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			20.10								1,100.00	
DL			0.10								10.00	

Domestic Well - Well #8

Well #8

Sample description: Well #8

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			14.10								925.00	
DL			0.10								10.00	

Domestic Well - Well #9

Domestic Well - Well #9

Well #9

Sample description: Well #9

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											3,010.00	
DL											10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 08/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										29.30	38
DL	0.10										10.00	10

D. SOIL ANALYSES

1

Field #1

Sample and source description: Field #1

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,706.00				
DL			1.00				

12-NE

12-NE

Sample and source description: 12-NE

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,616.00				
DL			1.00				

12-NW

12-NW

Sample and source description: 12-NW

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,616.00				
DL			1.00				

12-SE

12-SE

Sample and source description: 12-SE

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,616.00				
DL			2.00				

12-SW

12-SW

12-SW

Sample and source description: 12-SW

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,616.00				
DL			1.00				

13-E

13-E

Sample and source description: 13-E

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			862.00				
DL			1.00				

13-N

13-N

Sample and source description: 13-N

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			990.00				
DL			1.00				

13-S

13-S

13-S

Sample and source description: 13-S

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			980.00				
DL			1.00				

2

Field #2

Sample and source description: Field #2

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,294.00				
DL			1.00				

4

Field #4

Sample and source description: Field #4

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,935.00				
DL			1.00				

5

5

Field #5

Sample and source description: Field #5

Sample date: 05/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,325.00				
DL			1.00				

9

Field #9

Sample and source description: Field #9

Sample date: 08/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3,183.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

1 - 11/02/2022: Oats, silage-soft dough

1

Sample and source description: 1

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	3,500.00	26,500.00		10.00
DL	100.00	100.00	100.00		0.01

1 - 06/03/2023: Corn, silage

1 - 06/03/2023: Corn, silage

1

Sample and source description: 1

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,900.00	16,500.00		5.60
DL	100.00	100.00	100.00		0.01

11 - 11/15/2020: Alfalfa, hay

11

Sample and source description: 11

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	39,800.00	2,200.00	14,800.00		9.50
DL	100.00	100.00	100.00		0.01

11

Sample and source description: 11

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,900.00	2,400.00	11,400.00		6.30
DL	100.00	100.00	100.00		0.01

12-Brad - 11/15/2020: Alfalfa, hay

12-Brad - 11/15/2020: Alfalfa, hay

12-Brad

Sample and source description: 12-Brad

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,500.00	2,300.00	12,000.00		6.00
DL	100.00	100.00	100.00		0.01

12-BRAD

Sample and source description: 12-BRAD

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	1,400.00	8,400.00		6.10
DL	100.00	100.00	100.00		0.01

12-NE - 11/01/2022: Oats, silage-soft dough

12-NE

Sample and source description: 12-NE

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	3,800.00	30,100.00		10.30
DL	100.00	100.00	100.00		0.01

12-NE - 06/02/2023: Corn, silage

12-NE - 06/02/2023: Corn, silage

12-NE

Sample and source description: 12-NE

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,100.00	16,300.00		5.60
DL	100.00	100.00	100.00		0.01

12-NW - 11/01/2022: Oats, silage-soft dough

12-NW

Sample and source description: 12-NW

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	3,600.00	28,600.00		9.80
DL	100.00	100.00	100.00		0.01

12-NW - 06/02/2023: Corn, silage

12-NW

Sample and source description: 12-NW

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	1,700.00	16,100.00		5.70
DL	100.00	100.00	100.00		0.01

12-SE - 11/01/2022: Oats, silage-soft dough

12-SE

Sample and source description: 12-SE

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	3,300.00	27,500.00		10.80
DL	100.00	100.00	100.00		0.01

12-SE - 06/02/2023: Corn, silage

12-SE

Sample and source description: 12-SE

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,900.00	16,100.00		5.40
DL	100.00	100.00	100.00		0.01

12-SW - 11/01/2022: Oats, silage-soft dough

12-SW

Sample and source description: 12-SW

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	3,600.00	29,500.00		10.70
DL	100.00	100.00	100.00		0.01

12-SW - 06/02/2023: Corn, silage

12-SW

Sample and source description: 12-SW

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,200.00	17,300.00		5.70
DL	100.00	100.00	100.00		0.01

13-E - 10/31/2022: Oats, silage-soft dough

13-E

Sample and source description: 13-E

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,200.00	3,300.00	26,900.00		9.70
DL	100.00	100.00	100.00		0.01

13-E - 06/01/2023: Corn, silage

13-E

Sample and source description: 13-E

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	2,200.00	16,200.00		5.30
DL	100.00	100.00	100.00		0.01

13-N - 10/31/2022: Oats, silage-soft dough

13-N

Sample and source description: 13-N

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	3,300.00	25,700.00		10.60
DL	100.00	100.00	100.00		0.01

13-N - 06/01/2023: Corn, silage

13-N

Sample and source description: 13-N

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	1,900.00	17,500.00		5.80
DL	100.00	100.00	100.00		0.01

13-S - 10/31/2022: Oats, silage-soft dough

13-S

Sample and source description: 13-S

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	3,700.00	28,100.00		9.70
DL	100.00	100.00	100.00		0.01

13-S - 06/01/2023: Corn, silage

13-S

Sample and source description: 13-S

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,800.00	15,500.00		5.80
DL	100.00	100.00	100.00		0.01

14 - 05/08/2022: Alfalfa, hay

14

Sample and source description: 14

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,600.00	2,300.00	17,000.00		9.30
DL	100.00	100.00	100.00		0.01

14

Sample and source description: 14

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	1,400.00	8,900.00		5.80
DL	100.00	100.00	100.00		0.01

16 - 10/31/2022: Oats, silage-soft dough

16 - 10/31/2022: Oats, silage-soft dough

16

Sample and source description: 16

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	2,800.00	23,400.00		9.30
DL	100.00	100.00	100.00		0.01

16 - 06/01/2023: Corn, silage

16

Sample and source description: 16

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,900.00	13,900.00		4.70
DL	100.00	100.00	100.00		0.01

17 - 05/08/2022: Alfalfa, hay

17

Sample and source description: 17

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,500.00	2,300.00	18,800.00		9.70
DL	100.00	100.00	100.00		0.01

17 - 05/08/2022: Alfalfa, hay

17

Sample and source description: 17

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,100.00	2,300.00	20,200.00		8.40
DL	100.00	100.00	100.00		0.01

2 - 11/02/2022: Oats, silage-soft dough

2

Sample and source description: 2

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	4,700.00	30,500.00		11.00
DL	100.00	100.00	100.00		0.01

2 - 06/03/2023: Corn, silage

2

Sample and source description: 2

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	1,500.00	18,200.00		5.90
DL	100.00	100.00	100.00		0.01

3 - 11/03/2022: Oats, silage-soft dough

3

Sample and source description: 3

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	3,200.00	25,500.00		9.80
DL	100.00	100.00	100.00		0.01

3 - 06/04/2023: Corn, silage

3

Sample and source description: 3

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	2,400.00	17,000.00		5.90
DL	100.00	100.00	100.00		0.01

4 - 11/03/2022: Oats, silage-soft dough

4

Sample and source description: 4

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	2,900.00	26,900.00		10.00
DL	100.00	100.00	100.00		0.01

4 - 06/04/2023: Corn, silage

4

Sample and source description: 4

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,100.00	15,600.00		5.30
DL	100.00	100.00	100.00		0.01

5 - 11/03/2022: Oats, silage-soft dough

5A

Sample and source description: 5A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	4,100.00	34,600.00		9.90
DL	100.00	100.00	100.00		0.01

5B

Sample and source description: 5B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,300.00	3,600.00	28,000.00		9.70
DL	100.00	100.00	100.00		0.01

5 - 06/04/2023: Corn, silage

5 - 06/04/2023: Corn, silage

5A

Sample and source description: 5A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,000.00	16,900.00		5.20
DL	100.00	100.00	100.00		0.01

5B

Sample and source description: 5B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	1,900.00	17,300.00		5.60
DL	100.00	100.00	100.00		0.01

6 - 04/30/2022: Alfalfa, hay

6

Sample and source description: 6

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	45,400.00	2,900.00	16,500.00		11.10
DL	100.00	100.00	100.00		0.01

6 - 04/30/2022: Alfalfa, hay

6

Sample and source description: 6

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	1,100.00	6,900.00		4.70
DL	100.00	100.00	100.00		0.01

7 - 11/03/2022: Oats, silage-soft dough

7

Sample and source description: 7

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	3,400.00	28,600.00		9.90
DL	100.00	100.00	100.00		0.01

7 - 06/04/2023: Corn, silage

7

Sample and source description: 7

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,500.00	15,200.00		5.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

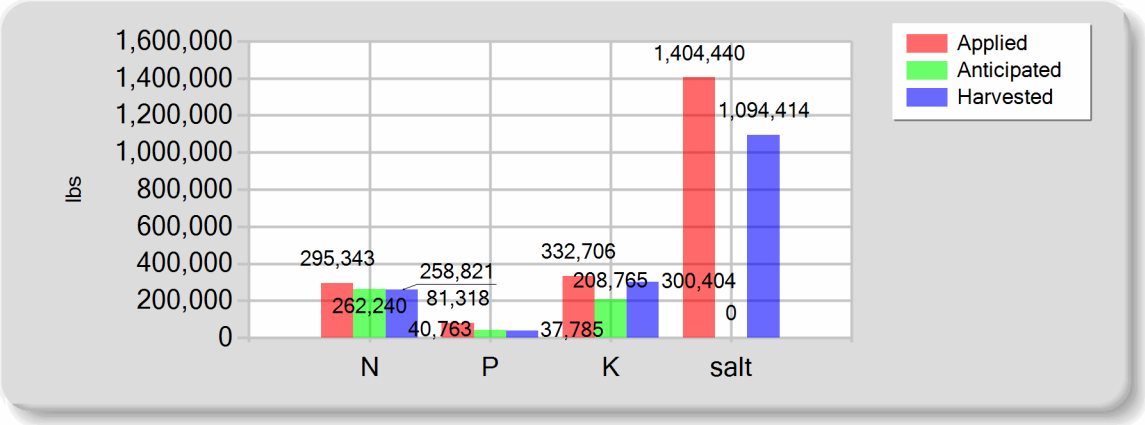
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

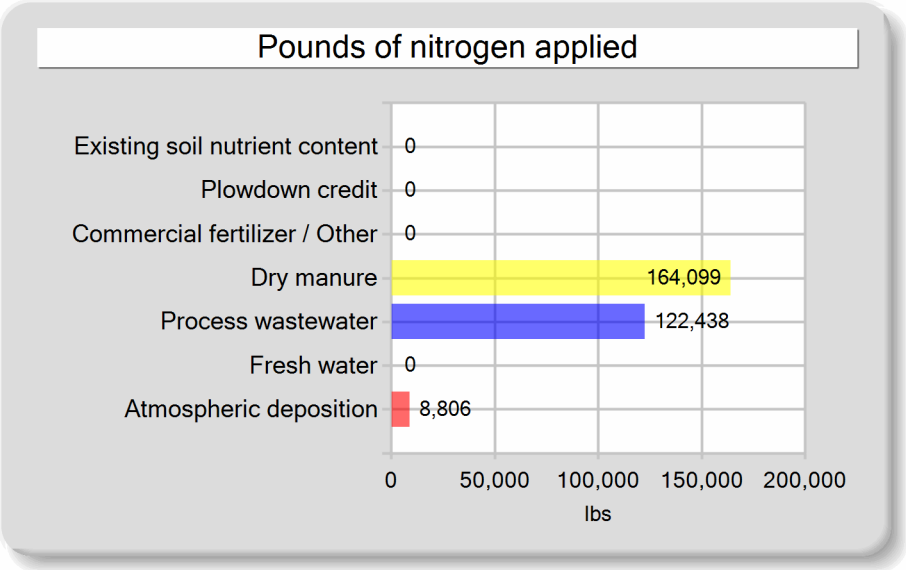
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

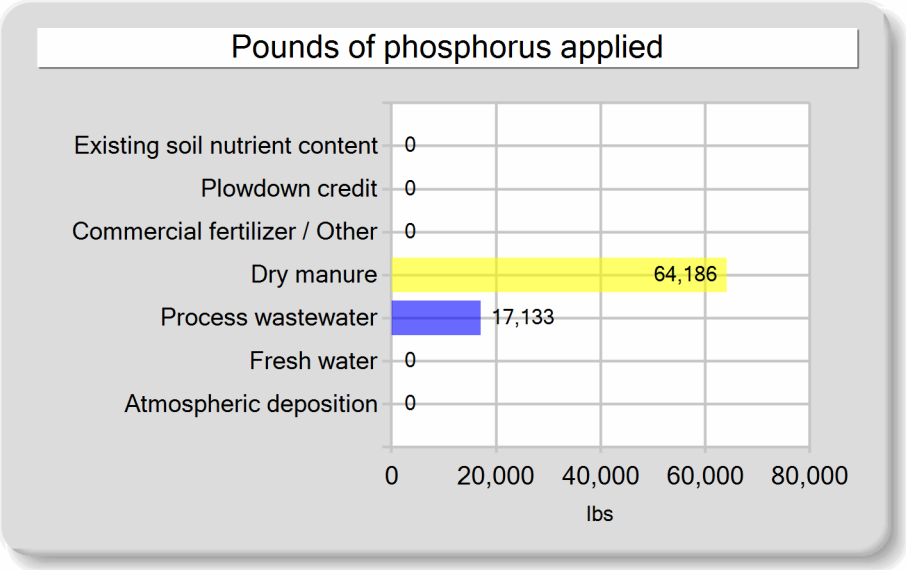
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164,098.97	64,185.55	110,445.31	0.00
Process wastewater	122,438.00	17,132.87	222,260.63	1,138,327.43
Fresh water	0.00	0.00	0.00	266,112.69
Atmospheric deposition	8,806.00	0.00	0.00	0.00
Total nutrients applied	295,342.97	81,318.42	332,705.94	1,404,440.12
Anticipated crop nutrient removal	262,240.00	40,763.20	208,764.80	0.00
Actual crop nutrient removal	258,820.67	37,785.47	300,404.24	1,094,413.74
Nutrient balance	36,522.30	43,532.94	32,301.70	310,026.38
Applied to removed ratio	1.14	2.15	1.11	1.28

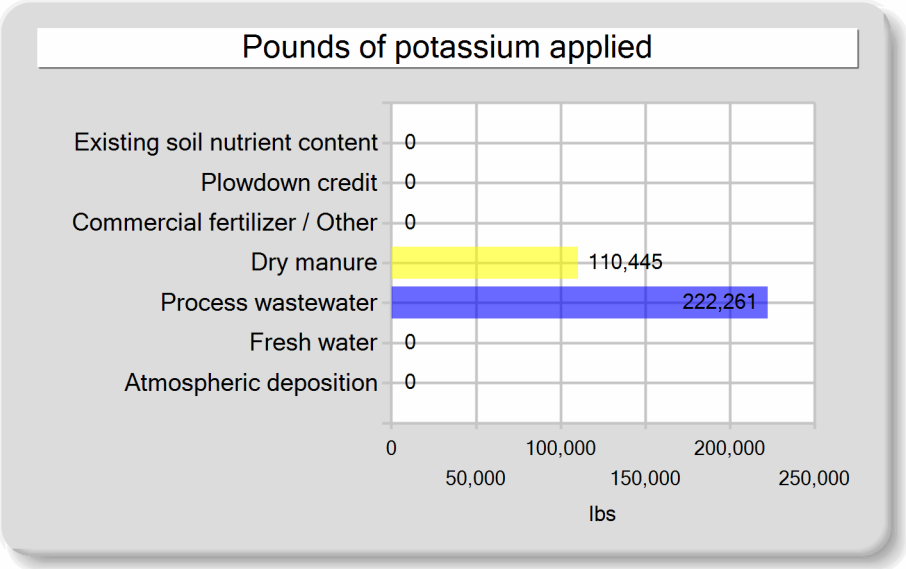
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

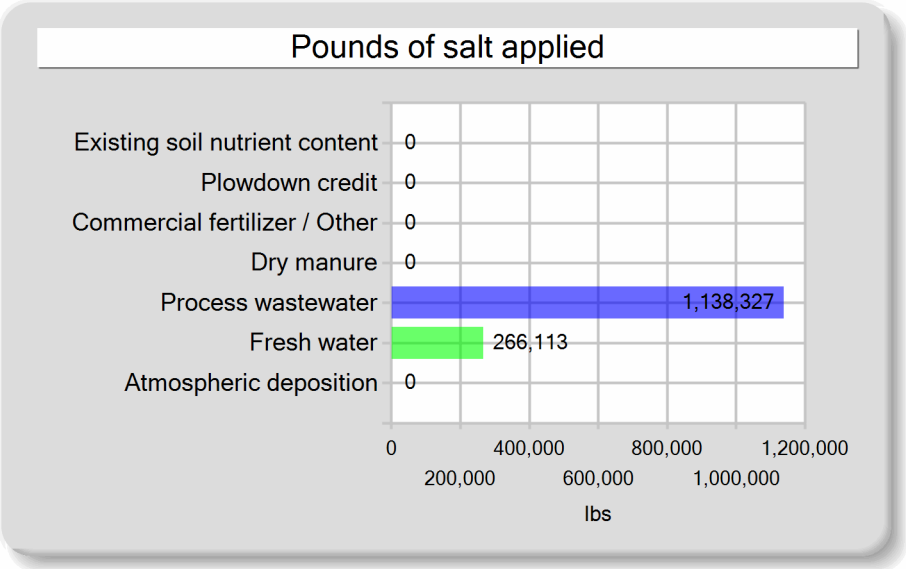


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

SOIL Sample values were entered as P2O5

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Hoffman Limited Partnership	S & S Dairy Inc.
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/19/2024	06/19/2024
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Darin Bylsma**

Name of Dairy Facility: **S & S Dairy, Inc.**

Facility Address: **384 E. Monte Vista Ave. Ceres, Ca. 95307**

Contact Person Name and Phone Number: **Darin Bylsma 209-606-4893**

Manure Hauler Information:

Name of Hauling Company/Person: **S & S Dairy, Inc.**

Address of Hauling Company /Person: **384 E. Monte Vista Ave. Ceres, Ca. 95307**

Contact Person and Phone Number: **Darin Bylsma 209-606-4893**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Oliveira Holsteins	**730 W. Monte Vista Ave.**	**Ceres**	**95307**	**209-595-6910**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
730 W. Monte Vista Ave	**Ceres**	**95307**	

Dates Hauled: **4/02/23, 5/22/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **N/A** Tons

Manure Solids Content (if amount reported in tons): N/A

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **800,000** Gallons

Method used to determine volume of process wastewater: _____

Pump Run Time x Average Pump GPM

Written Agreement:

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 06/19/2024

Hauler's Signature: _____ Date: 06/19/2024

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Darin Bylsma**

Name of Dairy Facility: **S & S Dairy, Inc.**

Facility Address: **384 E. Monte Vista Ave. Ceres, Ca. 95307**

Contact Person Name and Phone Number: **Darin Bylsma 209-606-4893**

Manure Hauler Information:

Name of Hauling Company/Person: **S & S Dairy, Inc.**

Address of Hauling Company /Person: **384 E. Monte Vista Ave. Ceres, Ca. 95307**

Contact Person and Phone Number: **Darin Bylsma 209-606-4893**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Oliveira Holsteins	**730 W. Monte Vista Ave.**	**Ceres**	**95307**	**209-595-6910**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
730 W. Monte Vista Ave	**Ceres**	**95307**	

Dates Hauled: **7/10/23, 7/19/23, 8/10/23 & 8/21/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **N/A** Tons

Manure Solids Content (if amount reported in tons): N/A

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **1,112,400** Gallons

Method used to determine volume of process wastewater: _____

Pump Run Time x Average Pump GPM

Written Agreement:

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 06/19/2024

Hauler's Signature: _____ Date: 06/19/2024

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K1074-01	Well #1	Ag Water	T. Madruga		11/20/2023 14:30
23K1074-02	Well #3	Ag Water	T. Madruga		11/20/2023 14:45
23K1074-03	Well #4	Ag Water	T. Madruga		11/20/2023 15:00
23K1074-04	Well #5	Ag Water	T. Madruga		11/20/2023 15:15
23K1074-05	Well #7	Ag Water	T. Madruga		11/20/2023 15:30
23K1074-06	Well #8	Ag Water	T. Madruga		11/20/2023 15:45
23K1074-07	Well #9	Ag Water	T. Madruga		11/20/2023 16:00

Default Cooler
Temperature on Receipt °C: -0.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results

Sample: Well #1
23K1074-01 (Water)

Sampled: 11/20/2023 14:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.62**	mmhos/cm	0.01	1		11/28/23 14:39	SM 2510 B		BEK0748
Electrical Conductivity umhos	**1620**	umhos/cm	10.0	1		11/28/23 14:39	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 14:30	Field		BEK0698
Nitrate Nitrogen as NO3N	**38.0**	mg/L	0.1	1	10	11/21/23 20:06	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:39	SM 4500-H+	H	BEK0748
pH	**7.5**	units	1.0	1		11/28/23 14:39	SM 4500-H+	H	BEK0748

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LABORATORY INC

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results
(Continued)

Sample: Well #3
23K1074-02 (Water)

Sampled: 11/20/2023 14:45
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.81**	mmhos/cm	0.01	1		11/28/23 14:40	SM 2510 B		BEK0748
Electrical Conductivity umhos	**1810**	umhos/cm	10.0	1		11/28/23 14:40	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 14:45	Field		BEK0698
Nitrate Nitrogen as NO3N	**69.8**	mg/L	0.1	1	10	11/21/23 20:25	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:40	SM 4500-H+	H	BEK0748
pH	**7.7**	units	1.0	1		11/28/23 14:40	SM 4500-H+	H	BEK0748

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results
(Continued)

Sample: Well #4
23K1074-03 (Water)

Sampled: 11/20/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.86**	mmhos/cm	0.01	1		11/28/23 14:41	SM 2510 B		BEK0748
Electrical Conductivity umhos	**1860**	umhos/cm	10.0	1		11/28/23 14:41	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 15:00	Field		BEK0698
Nitrate Nitrogen as NO3N	**71.7**	mg/L	0.1	1	10	11/21/23 20:45	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:41	SM 4500-H+	H	BEK0748
pH	**7.7**	units	1.0	1		11/28/23 14:41	SM 4500-H+	H	BEK0748

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results
(Continued)

Sample: Well #5
23K1074-04 (Water)

Sampled: 11/20/2023 15:15
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.83**	mmhos/cm	0.01	1		11/28/23 14:43	SM 2510 B		BEK0748
Electrical Conductivity umhos	**1830**	umhos/cm	10.0	1		11/28/23 14:43	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 15:15	Field		BEK0698
Nitrate Nitrogen as NO3N	**60.5**	mg/L	0.1	1	10	11/21/23 21:05	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:43	SM 4500-H+	H	BEK0748
pH	**7.5**	units	1.0	1		11/28/23 14:43	SM 4500-H+	H	BEK0748

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results
(Continued)

Sample: Well #7
23K1074-05 (Water)

Sampled: 11/20/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.10**	mmhos/cm	0.01	1		11/28/23 14:44	SM 2510 B		BEK0748
Electrical Conductivity umhos	**1100**	umhos/cm	10.0	1		11/28/23 14:44	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 15:30	Field		BEK0698
Nitrate Nitrogen as NO3N	**20.1**	mg/L	0.1	1	10	11/21/23 21:24	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:44	SM 4500-H+	H	BEK0748
pH	**7.9**	units	1.0	1		11/28/23 14:44	SM 4500-H+	H	BEK0748

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results
(Continued)

Sample: Well #8
23K1074-06 (Water)

Sampled: 11/20/2023 15:45
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.93**	mmhos/cm	0.01	1		11/28/23 14:45	SM 2510 B		BEK0748
Electrical Conductivity umhos	**925**	umhos/cm	10.0	1		11/28/23 14:45	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 15:45	Field		BEK0698
Nitrate Nitrogen as NO3N	**14.1**	mg/L	0.1	1	10	11/21/23 21:44	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:45	SM 4500-H+	H	BEK0748
pH	**8.0**	units	1.0	1		11/28/23 14:45	SM 4500-H+	H	BEK0748

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Sample Results
(Continued)

Sample: Well #9
23K1074-07 (Water)

Sampled: 11/20/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**3.01**	mmhos/cm	0.01	1		11/28/23 14:46	SM 2510 B		BEK0748
Electrical Conductivity umhos	**3010**	umhos/cm	10.0	1		11/28/23 14:46	SM 2510 B		BEK0748
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 16:00	Field		BEK0698
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/21/23 22:04	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 14:46	SM 4500-H+	H	BEK0748
pH	**8.0**	units	1.0	1		11/28/23 14:46	SM 4500-H+	H	BEK0748

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: S & S Dairy

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:20

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0692										
Blank (BEK0692-BLK1)					Prepared & Analyzed: 11/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0692-BLK2)					Prepared & Analyzed: 11/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0692-BLK3)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0692-BS1)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		105	90-110		
LCS (BEK0692-BS2)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		106	90-110		
Duplicate (BEK0692-DUP1)		**Source: 23K1075-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	0.1		0.1	mg/L		0.1			2.76	10
Duplicate (BEK0692-DUP2)		**Source: 23K1076-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L		5.1			0.452	10
Matrix Spike (BEK0692-MS1)		**Source: 23K1075-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.1	101	90-110		
Matrix Spike (BEK0692-MS2)		**Source: 23K1076-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	10.3		0.1	mg/L	5.000	5.1	103	90-110		
Reference (BEK0692-SRM1)					Prepared & Analyzed: 11/21/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0692-SRM2)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0692-SRM3)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0748										
Blank (BEK0748-BLK1)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEK0748-BLK2)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0748-BLK3)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.4		1.0	units						
Duplicate (BEK0748-DUP1)		**Source: 23K1055-01**			Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	0.57		0.01	mmhos/cm		0.57			0.0698	10
Electrical Conductivity umhos	573		10.0	umhos/cm		573			0.0698	10
pH	5.0		1.0	units		5.0			0.00	10
Duplicate (BEK0748-DUP2)		**Source: 23K1075-03**			Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1.18		0.01	mmhos/cm		1.18			0.313	10
pH	8.1		1.0	units		8.1			0.123	10
Electrical Conductivity umhos	1180		10.0	umhos/cm		1180			0.313	10
Reference (BEK0748-SRM1)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	552			umhos/cm	538.0		103	90-110		
Reference (BEK0748-SRM2)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0748-SRM3)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0748-SRM4)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEK0748-SRM5)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		

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Received: 11/21/2023 16:09
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0748 (Continued)										
Reference (BEK0748-SRM5)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEK0748-SRM6)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0748-SRM7)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0748-SRM8)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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11/21/23 16:09 23K1074

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 7 No. of Bottles:

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT S & S Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well #1	11/20/2023	1430	-0.6	ND
2	Well #3	11/20/2023	1445	-0.9	ND
3	Well #4	11/20/2023	1500	-0.8	ND
4	Well #5	11/20/2023	1515	-0.6	ND
5	Well #7	11/20/2023	1530	-0.1	ND
6	Well #8	11/20/2023	1545	-0.6	ND
7	Well #9	11/20/2023	1600	1.0	ND
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/20/2023 @ 1600	11/21/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/21/2023 @ 900	11-21-23 3:48
Third				
Fourth		DLI	11/21/23 16:09	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/21/23 16:09 23K1074

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice:☒ Blue Ice:☐ None:☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1	1	1	1	1			
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										